Tracing Banks' Credit Allocation to their Profits*

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November 5, 2026

Abstract

We quantify how banks adjust their lending behavior following a shock to their inframarginal profits. For identification, we exploit banks' heterogeneous liability composition and the existence of regulated deposits in France whose rates are set by the government. Using administrative credit-registry and bank data, we find that a one-standard-deviation increase in the cost of regulated deposits relative to market rates (38.5 basis points) reduces credit supply by 0.87% at a bank with the average regulated-deposit ratio, with a response that is muted at low levels of this gap and grows as the gap widens. To insulate their profits, affected banks also reach for yield and rebalance their lending toward smaller and riskier firms. These changes are not compensated for by less affected banks at the aggregate city level, and as a result, firms for which we observe accounting data have to reduce their investment.

JEL classification: E21, G20, G21, G28

Keywords: bank funding costs, deposits, credit supply, SMEs, savings

Replication package: https://github.com/amatray/tracing-banks-credit-allocation

*We thank Cynthia Balloch, Christoph Basten, Tobias Berg, Lamont Black, Matteo Crosignani, Husnu Dalgic, Florian Heider, Victoria Ivashina, Arvind Krishnamurthy, Christian Kubitza, Frédéric Malherbe, Daniel Neuhann, Daniel Paravisini, Andrea Passalacqua, Diane Pierret, Philipp Schnabl, Philip Strahan, Javier Suárez, David Thesmar and Emily Williams, as well as seminar participants at the Boston Fed, University of Bonn, Frankfurt School of Finance & Management, ESMT Berlin, Banca d'Italia, Banque de France, CSEF Naples, Vienna Graduate School of Finance, the CU Boulder Finance in the Cloud Conference, the 2020 EFA Annual Meeting, the 2020 EABCN conference on "Empirical Advances in Monetary Policy," the 2020 Fixed Income and Financial Institutions Conference, the 2021 MFA Annual Meeting, the 10th MoFiR Workshop on Banking, the 2022 AFA Annual Meeting, the 2022 BSE Summer Forum, the 2023 Banque de France – TSE Workshop, the 2023 New York Fed/NYU Stern Conference on Financial Intermediation, and the 2024 AEA Annual Meeting for helpful comments. Saidi gratefully acknowledges funding by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy (EXC 2126/1 – 390838866) and through CRC TR 224 (Project C03). The views expressed do not necessarily reflect those of the European Central Bank, Federal Reserve Bank of Atlanta, Banque de France, the Eurosystem, or the Federal Reserve System.

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1 Introduction

How banks' funding and operating costs affect their behavior and are transmitted to the real economy is at the core of policy debates about the financial system, from the effectiveness of monetary policy to the effects of micro- and macroprudential regulations.[1] Banks' profits determine their net worth and ultimately the tightness of their financing constraints. Therefore, shocks to banks' profits, even if the latter are inframarginal, can affect their credit supply (e.g., Gertler and Kiyotaki, 2010) and their risk-taking incentives. However, as banks jointly optimize their assets and liabilities, identifying exogenous variation in such costs that does not directly affect the level of banks' liabilities or the profitability of their potential investments is challenging.

In this paper, we use variation that relates only to the cost of a funding source that is inframarginal to banks. This enables us to isolate a specific cost shifter and quantify its pass-through to the quantity and composition of banks' credit supply. To this end, we use rich administrative data over the period 2010–2015 in France, covering banks' balance-sheet information and a detailed breakdown of their funding structure, the near universe of bank loans to firms, as well as firms' balance sheets and income statements from tax returns.

For identification, we exploit the existence of regulated-deposit accounts offered to households in France. Unlike regular savings accounts, the rate on regulated deposits is neither determined by the banks themselves nor directly dependent on the monetary-policy rate. It is instead set by the government up to twice a year, following a formula indexed to money-market rates and inflation from which the government frequently deviates. These shifts in the cost of regulated deposits move independently of the monetary-policy rate and, hence, of the cost of banks' alternative funding sources.

Since, in addition, balance-sheet exposure to regulated deposits is stable over time and varies primarily across banks due to regulatory obligations, the rate on regulated deposits is an inframarginal cost. As such, fluctuations in this rate can be interpreted as shifts in

[1] The transmission of monetary policy to the real sector and its effectiveness depend in part on the pass-through to banks' funding costs (Gertler and Kiyotaki, 2010). More generally, banks' cost of capital can affect the quantity and quality of credit supply in response to both microprudential (Repullo and Suárez, 2013; Begenau, 2020) and macroprudential regulations (Jiménez, Ongena, Peydró, and Saurina, 2017).

the cost of maintaining a deposit franchise, allowing us to test to what extent such costs matter for bank behavior. A higher rate on regulated deposits cuts into banks' profits, which ultimately depletes their net worth and can influence their lending behavior along two margins: (i) a reduction of their credit supply and (ii) a reallocation of their credit portfolio toward higher-yielding loans so as to insulate their profits. This second margin, for which we provide evidence, is potentially of central importance for the transmission of credit shocks to economic growth since the reallocation of capital among heterogeneous producers can affect aggregate output, even if the net supply of total credit is unchanged (Bau and Matray, 2023; Baqaee, Farhi, and Sangani, 2024).

By comparing banks with a higher share of regulated deposits with otherwise-funded banks, we trace out the effects of these shifts in bank funding costs at different levels of aggregation: the bank-firm level, the firm level, and the city level. Because our measure of exposure exploits differences in the *composition* of bank liabilities, we can control for confounding effects that may be due to differences in the *level* of bank liabilities or leverage. As a result, we estimate the effect of a change in the cost of funding net of any change in total liabilities (such as fluctuations in deposits as in Drechsler, Savov, and Schnabl, 2017; Begenau and Stafford, 2025) or in the returns on their assets that could directly affect bank lending.

Our first set of results shows how banks' credit supply responds to regulatory-driven variation in funding costs. A one-standard-deviation increase in Gap_t, the difference between the regulated-deposit rate and market rates (38.5 basis points), reduces lending by 0.87% at a bank with the average regulated-deposit ratio of 14%. This average hides a strong nonlinearity: relative to quarters in which Gap_t is at most 125 basis points, the coefficient grows in magnitude from -0.097 when Gap_t is between 125 and 150 basis points to -0.181 when it is between 175 and 200 basis points. Per percentage point of banks' average funding costs, the same estimate implies a 16% contraction in lending. The cost differences due to banks' reliance on regulated deposits are equivalent to a shift in banks' total fixed costs of operation or their investment in a deposit franchise, which in turn translates into fluctuations in bank profits and, ultimately, equity. This allows us to back out the sensitivity of banks'

credit supply to their profitability: an increase in banks' operating expenses that depresses their equity by 1% translates to a reduction in their credit supply by 0.67%.

We obtain our estimate using granular data on loans at the bank-firm-time level that allow us to implement a standard within-firm estimator to control for any changes in firms' credit demand that might be correlated with banks' funding costs.

While we control for many time-varying elements that are potentially correlated with bank lending, we cannot, by definition, account for time-varying *unobserved* heterogeneity across banks. We address the possibility that time-varying bank-level characteristics may be correlated with changes in the relative cost of regulated deposits in several ways.

First, we include a battery of high-dimensional fixed effects such as banks' county-by-time and banking group (BHC)-by-time fixed effects. County-by-time fixed effects ensure that we only exploit variation across banks in the *same* county, so that lending decisions cannot be affected by differences in local market power or local business cycles. The inclusion of BHC-by-time fixed effects, in turn, implies that we use variation across banks belonging to the same group, thereby netting out any differences in top-management styles and abilities, the impact of regulation, or broader funding shocks such as a run on the wholesale funding market.

Second, our results are similar when we use only variation in the composition of total deposits rather than total liabilities, and when we augment our specification to include bank-level covariates interacted with the cost shifter. This implies that changes in lending behavior are driven neither by time-varying factors correlated with banks' general dependence on deposit funding (such as their business models) nor by aggregate shocks that would have heterogeneous effects on specialized banks.

We also explore how the elasticity of credit supply to changes in the relative cost of regulated deposits varies across banks. Lending drops by more for weakly capitalized banks, and for banks with lower liquidity buffers to absorb the cost increase. This points to an amplification of the sensitivity of banks' credit supply if their probability of default is higher, consistent with higher average funding costs affecting banks' credit supply via a change in their expected net worth (e.g., Gertler and Kiyotaki, 2010; Bahaj and Malherbe, 2020).

Our second set of results shows that a key margin of adjustment for banks is not only the net credit supply but also the change in credit composition. Banks rebalance their loan portfolios across borrowers and loan characteristics in an effort to shield their profits from funding-cost fluctuations. When the cost of regulated deposits increases, affected banks engage in relatively greater risk taking and shift their portfolios toward higher-yielding loans. They do so by increasing the average maturity of their loans and their exposure to riskier firms, such as smaller firms or firms operating in industries with higher bankruptcy risk.

The magnitude of this credit reallocation is sizable. Between 55% and 72% of the decline in credit to large firms at affected banks reflects a shift of their portfolios toward small, and arguably riskier, firms rather than a proportional contraction of lending. At the firm level, we show that this reallocation is large enough to generate a bigger credit contraction among larger firms and firms with longer main-bank relationships, despite the fact that these firms are usually thought best able to switch lenders. This implies that focusing on changes in net credit supply underestimates the true effect of bank-level shocks on the real economy since, in an attempt to insulate their profits, banks reallocate substantial amounts of capital across heterogeneous producers in addition to contracting their credit supply.

In our third set of results, we use the methodology of Beaumont, Matray, Tu, and Xu (2026) to show that credit does not reallocate across space, as the negative credit-supply shock is constant when we allow for different levels of geographic spillovers. The implications are twofold. First, banks less reliant on regulated deposits do not step in to serve the unaddressed local loan demand, potentially because lending relationships are sticky. Second, this opens up the possibility that variation in banks' funding costs has real economic effects, e.g., on firm-level investment, at least for those firms that are adversely affected by banks' lending decisions in the face of higher funding costs.

To test this, we aggregate our bank-level shock at the firm level using the loan exposure of firms to each of their existing lenders. We find that firms more exposed to regulated-deposit dependent banks reduce their tangible assets and stock of total capital assets when the relative cost of regulated deposits increases.

Relation to literature. Identified shocks to banks' credit supply typically originate on the liability side through an interbank liquidity drought (Iyer, Peydró, da-Rocha-Lopes, and Schoar, 2013; Cingano, Manaresi, and Sette, 2016; De Jonghe, Dewachter, Mulier, Ongena, and Schepens, 2019), a deposit outflow (Drechsler, Savov, and Schnabl, 2017), or a change in capital requirements (Jiménez, Ongena, Peydró, and Saurina, 2017; Gropp, Mosk, Ongena, and Wix, 2019), and therefore move three things at once: the bank's quantity of available funding, its marginal cost of funding, and its net worth. Our setting moves only the last.

The government-set rate on regulated deposits changes the price of a liability whose quantity banks cannot adjust, making it inframarginal to their lending decision, while leaving unchanged both the level of liabilities and their composition between deposits and wholesale funding. This also allows us to hold constant the business-model differences that separate deposit-funded from market-funded banks and would otherwise confound the liability-side shock with bank-specific lending behavior (Paravisini, Rappoport, and Schnabl, 2023). Our setting therefore lets us isolate a shock to bank profits, and cumulatively to bank net worth, that is unrelated to the asset side. Unbundling the net-worth channel from the funding-quantity and marginal-cost channels allows us to speak to four questions that cannot be fully addressed with the shocks studied so far in the banking literature.

First, we isolate the net-worth, or bank-capital, channel of credit supply from the lending channel. The former operates through a higher external-finance premium when a bank's net worth falls (Gertler and Kiyotaki, 2010; Bahaj and Malherbe, 2020); the latter through the quantity of loanable funds (Kashyap and Stein, 2000; Kishan and Opiela, 2000; Jiménez, Ongena, Peydró, and Saurina, 2012), and any funding-quantity shock shifts both. We show that a pure profit shock with unchanged deposit quantities reduces lending, particularly so at weakly capitalized and illiquid banks. The effect is muted for small cost increases and grows with their size, which is what a convex cost of external finance predicts. This curvature cannot be recovered from the one-time shocks on which the literature identifying credit-supply shocks and their real effects mostly relies (e.g., Peek and Rosengren, 2000; Khwaja and Mian, 2008; Paravisini, 2008; Becker and Ivashina, 2014; Chodorow-Reich, 2014; Huber, 2018; Carlson, Correia, and Luck, 2022; Xu, 2022), nor from capital-requirement

experiments, in which banks respond to a regulatory ratio rather than to their net worth (Behn, Haselmann, and Wachtel, 2016; Gropp, Mosk, Ongena, and Wix, 2019).

Second, we provide a direct test in the debate over the effect of banks' deposit franchise on their lending behavior. Drechsler, Savov, and Schnabl (2021) argues that the franchise (a sticky, low-rate deposit base maintained at a fixed operating cost) hedges banks' interest rate risk and stabilizes their net interest margins; Begenau and Stafford (2022) counters that stable margins are uninformative about hedging; and d'Avernas, Eisfeldt, Huang, Stanton, and Wallace (2023) shows that large and small banks run different deposit business models. In franchise-value models, the franchise is worth the present value of deposit spreads net of the cost of operating it (Jiang, Matvos, Piskorski, and Seru, 2024; Bolton, Li, Wang, and Yang, 2025; Drechsler, Savov, Schnabl, and Wang, 2026). A government-set rate on a quantity-fixed deposit base is a shock to that cost. We find that it passes through to credit supply (the operating cost of the franchise affects lending as well as margins) and that banks respond by rebalancing their loan portfolios toward higher-yielding, longer-maturity, and riskier loans. This asset-side response suggests that the observed stability of net interest margins may in part reflect active portfolio adjustment rather than liability-side hedging, which is one concrete mechanism behind the concern raised by Begenau and Stafford (2022).

Third, because the gap between the livret-A rate and the policy rate is uncorrelated with the policy rate and moves in both directions, we can separate the profitability channel of bank risk taking from the level of interest rates. Theory predicts that banks whose intermediation margins are compressed search for yield (Martinez-Miera and Repullo, 2017), and evidence from low- and negative-rate environments finds more risk taking by the deposit-reliant banks whose margins are squeezed (Heider, Saidi, and Schepens, 2019; Bubeck, Maddaloni, and Peydró, 2020; Whited, Wu, and Xiao, 2021; Eggertsson, Juelsrud, Summers, and Wold, 2024). In those settings, however, the squeeze is itself caused by low rates, which at the same time lower banks' marginal cost of funding (the leverage-based risk-taking channel of monetary policy, Jiménez, Ongena, Peydró, and Saurina, 2014; Ioannidou, Ongena, and Peydró, 2015; Dell'Ariccia, Laeven, and Suarez, 2017; Paligorova and Santos, 2017) and the return on safe assets, the classic motive for reaching for yield (Hanson and Stein, 2015),

so these effects cannot be told apart. In our setting, the margin squeeze is orthogonal to the policy rate, so both marginal funding costs and the returns on alternative assets are held fixed, and banks still shift toward higher-yielding loans. This also contrasts with the mechanism in Drechsler, Savov, and Schnabl (2018), where a wider deposit spread raises the price of liquidity and banks reduce risk taking: we show that a higher operating cost of maintaining a deposit franchise can lead to more risk taking.

Fourth, our reallocation results show that a shock to bank net worth changes the direction of credit as well as its quantity. The literature on credit reallocation after bank funding shocks finds that banks retreat toward borrowers and sectors in which they are specialized or hold stronger relationships (De Jonghe, Dewachter, Mulier, Ongena, and Schepens, 2019; Paravisini, Rappoport, and Schnabl, 2023). A pure profit shock produces a different pattern: most of the decline in credit to large firms at affected banks reflects a shift of their portfolios toward small and riskier firms, and these banks also lengthen loan maturities and increase exposure to high-bankruptcy industries, consistent with an attempt to restore profits rather than to preserve expertise. Such reallocation matters because, with heterogeneous borrowers, it affects aggregate output even at a constant total volume of credit (Bau and Matray, 2023; Baqaee, Farhi, and Sangani, 2024), so the net credit response alone understates the real effects of shocks to bank profitability.

The compositional response we document survives at the city level, where total credit falls by slightly more than at the bank-firm level, indicating that unaffected lenders do not absorb the displaced borrowers. This differs from Jiménez, Mian, Peydró, and Saurina (2020), which shows that the large loan-level effect of a securitization-driven credit expansion is neutralized at the firm level, as multi-bank borrowers offset it by reducing credit from other banks. The difference reflects the sign of the shock and the state of the credit market: an expansion by one lender crowds out others because firm demand is finite, whereas a contraction requires another lender to extend new credit to a borrower it knows less well (Darmouni, 2020), which is harder under the normal business-cycle conditions of our sample than in a liquidity-abundant boom.

Taken together, the most surprising implication of our results is that an inframarginal

cost moves the marginal lending decision. In a frictionless world, the cost of a liability whose quantity a bank cannot adjust should not affect the loan it makes at the margin; we find that it does, that the effect is large, and that banks respond to the resulting profit squeeze by taking more risk rather than less. This has direct implications for policy. Any measure that raises the fixed cost of maintaining a deposit base (e.g., bank levies or compliance costs) may affect the volume and the composition of credit, whether or not it is intended to do so. Our estimates provide the relevant elasticities: a measure that reduces bank equity by 1% lowers credit supply by about 0.7%, with the burden falling on credit to larger firms, while measures too small to substantially affect average funding costs are absorbed without a lending response. The nonlinearity also means that the aggregate cost of such measures depends on their size and on the capitalization of the banks they fall on.

Finally, to approximate the funding costs of banks that do not rely on regulated deposits, we use the pass-through of the policy rate to other deposit and market rates (Hannan and Berger, 1991; Driscoll and Judson, 2013; Drechsler, Savov, and Schnabl, 2017). This connects our paper to the literature on heterogeneous monetary transmission through banks' funding structure, whether due to imperfect competition for deposits (Drechsler, Savov, and Schnabl, 2017; 2021; Balloch and Koby, 2022; Wang, Whited, Wu, and Xiao, 2022) or a lower bound on retail deposit rates (Heider, Saidi, and Schepens, 2019; Bubeck, Maddaloni, and Peydró, 2020; Eggertsson, Juelsrud, Summers, and Wold, 2024), both of which lower the rate sensitivity of banks' liabilities relative to their assets (Gomez, Landier, Sraer, and Thesmar, 2021). In all of these settings, however, deposit rates are set by banks themselves. Our paper identifies sticky deposit rates that are not the outcome of bank price setting, and uses them as a source of variation in funding costs that is exogenous to banks' decisions.

2 Background and Empirical Strategy

2.1 Regulated-deposit Accounts in France

By the end of 2021, regulated deposits accounted for 14% of French households' total financial assets. As they are risk-free, tax-free, highly liquid, and have a very low entry

threshold (minimum of €15), these accounts are the most popular savings scheme in France for medium- and low-income households subject to income tax. Most importantly, regulated deposits pay interest at a rate set by the government that banks cannot adjust.

2.1.1 Livret A

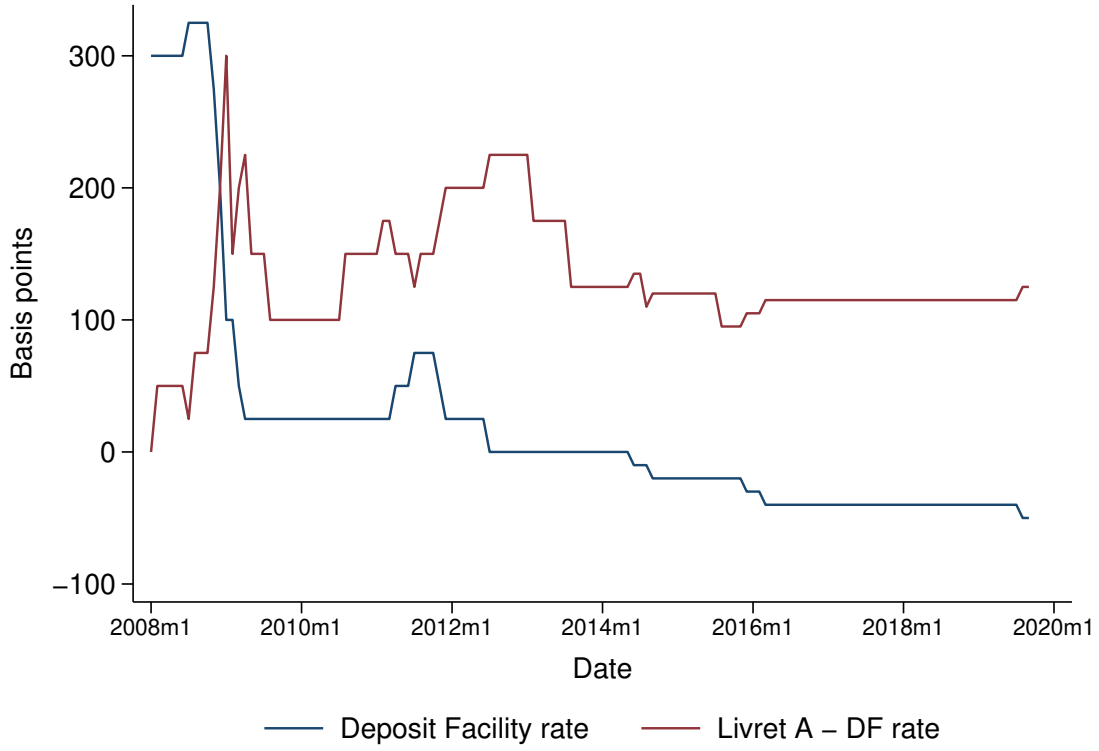
The most common regulated-deposit account is called "livret A," which can be opened by any individual or non-profit organization. It was established in 1818 to pay back the debts incurred during the Napoleonic wars, and was originally distributed by three "incumbent" banks (La Banque Postale, Caisses d'Epargne et de Prévoyance, and Crédit Mutuel). The Law of Modernization of the Economy extended the right to offer livret-A accounts to all French credit institutions (including "new banks") starting January 1, 2009. In spite of the rates being set by the government, French banks widely offer such accounts because French depositors have a strong preference for them and tend to fill their regulated-deposit accounts up to the cap before demanding any regular savings products and other, non-savings products. Banks, therefore, are de facto forced to offer regulated deposits to be competitive.

Given the popularity of livret-A accounts, the government had to impose a cap, often binding for middle-income households, on how much money can be saved in this form. Each depositor can only hold a single livret A, and deposits cannot exceed €22,950 for individuals (not including capitalized interest) or €76,500 for non-profit legal entities.[2] Regulated deposits include livret A, which represents one-third of such deposits, as well as other types of savings accounts for which the rates are pegged to the livret-A rate. The rate is the same as, or above, the livret-A rate for most of these regulated deposits (LDD, Livret Jeunes, LEP, PEL), and is equal to two-thirds of the livret-A rate for one type of account (CEL), which only accounts for 5% of total regulated deposits.

The livret-A rate is set by the government. It is calculated by the French Central Bank twice a year, on January 15 and July 15, and becomes effective on February 1 and August 1, respectively. Over our sample period from 2010 to 2015, the formula sets the livret-A rate

[2] After the financial crisis and the European sovereign debt crisis, this product was so popular that the government increased the maximum amount by 50%, in two stages, from €15,300 to €19,125 and €22,950 in October 2012 and January 2013, respectively.

Figure 1: **Changes in Funding-cost Gap.** This figure shows the evolution of the ECB's deposit facility (DF) rate and the gap between the livret-A rate and the latter from 2008 to 2019.



to the higher of two terms:

$$r_t^A = \max\left\{\frac{1}{2}\left(\frac{\text{Euribor}_t^{3M} + \text{Eonia}_t}{2} + \pi_t\right),\ \pi_t + 0.25\right\},$$

where Euribor_t^{3M} and Eonia_t are the monthly averages of the three-month Euribor rate and the euro overnight index average rate, and π_t is the French inflation rate, measured as the percentage change in the consumer price index over the latest available 12 months, all in percent. The government can deviate from this revision procedure and has the discretion to decide a new rate, which has been very common in practice.[3] The reasons stated for these deviations concern household savers rather than bank lending. In February 2012, the

[3] For instance, on February 1, 2012, the Minister of the Economy, François Baroin, following the recommendation of the Governor of the French Central Bank, decided to maintain the rate at 2.25%, although the inflation rate would have prompted an increase in the livret-A rate to 2.75%. On February 1, 2013, the Minister of the Economy at the time, Pierre Moscovici, lowered the livret-A rate only to 1.75% when the strict application of the formula would have implied a greater drop, to 1.5%. Similarly, on August 1, 2013, the livret-A rate was reduced to 1.25% instead of 1%. And on February 1, 2014, although the Governor of the French Central Bank recommended lowering the rate to 1%, and the formula actually implied lowering it further to 0.75%, the Minister decided to keep the livret-A rate at 1.25%.

Governor of the French Central Bank recommended not applying the formula because he expected inflation to fall, and the Minister of the Economy followed this recommendation. In January 2013 and January 2014, the Minister invoked savers' purchasing power, in January 2014 against the Governor's recommendation. In July 2013, the Governor himself proposed departing from the formula, because the cut it implied would have brought the rate to an unprecedented low.[4] Since these reasons refer to current or expected inflation, the deviations are not independent of macroeconomic conditions. In column 10 of Table 5, we compare the lending response to changes in the rate that follow the formula with the response to changes that stem from the government's political deviations from the formula.

Thus, unlike rates on ordinary savings accounts or interbank funding, the rate on regulated deposits does not track the monetary-policy rate and fluctuates for reasons independent of it. In Figure 1, we plot the time-series variation in the difference between the livret-A rate and the main policy rate of the European Central Bank (ECB), the deposit facility rate.[5] From 2010 to 2014, the ECB's monetary policy is both contractionary and expansionary, whereas the difference between the livret-A and the deposit facility rate tends to increase over the same time period. The correlation between this difference and the actual monetary-policy rate during this period is -0.01.

2.1.2 Banks' Funding Costs and Credit Supply

The existence of regulated deposits in France allows us to exploit changes in banks' profits stemming from the interaction between the rates on regulated deposits and the amount of these deposits among banks' liabilities, both of which vary for reasons largely independent of banks' decisions, as we explain below.

Rates on regulated deposits. These rates cannot be adjusted by banks, but are set by the government, following a formula with frequent political deviations from it, and move

[4] See the announcements reported by La finance pour tous on February 9, 2012, July 12, 2013, and January 15, 2014, and by France Info on January 15, 2013. Appendix C reproduces the relevant passage of each announcement in English translation, with its source.

[5] See Figure B.1 in the Online Appendix for a version of the figure that separately plots the livret-A rate and the ECB's deposit facility rate.

Table 1: Evolution of Percentage of Eligible Regulated Deposits Transferred to the CDC

	2010	2011	2012	2013	2014	2015
Incumbent banks (prior to the reform in 2008)	80%	76%	70%	64%	62%	61%
New banks	24%	34%	40%	37%	40%	40%

Notes: Source: Regulated Savings Observatory of the Banque de France (Observatoire de l'épargne réglementée).

independently of the monetary-policy rate (as exemplified by the low correlation between the monetary-policy rate and the difference between the latter and the livret-A rate in Figure 1). Besides regulated deposits, banks fund themselves by issuing other deposits or through the interbank market. Compared with the livret-A rate, the rates on these alternative funding sources are significantly more aligned with the monetary-policy rate: retail deposit rates exhibit primarily upward, but not downward, stickiness, and interbank rates track the monetary-policy rate in the euro area relatively well despite higher post-crisis liquidity and counterparty risk (e.g., Illes and Lombardi, 2013; Heider, Saidi, and Schepens, 2019). This typical strong pass-through of the monetary-policy rate allows us to use the latter to approximate the cost for the portion of bank funding that does not come from regulated deposits.

Amounts of regulated deposits. The amount of regulated deposits on a bank's balance sheet is largely outside banks' control, for two reasons. First, as explained in Section 2.1.1, regulated deposits are in high demand by households but could initially only be offered by certain banks, which created a strong path dependence in market shares. Second, by law, banks retain on their balance sheet only a fraction of the regulated deposits that they collect.

A large portion of the collected savings is rechanneled to a special fund operated by a state-owned financial institution, the Caisse des Dépôts et Consignations (CDC), to finance social housing. Only a subset of regulated deposits is rechanneled to the CDC. We refer to them as eligible deposits, of which livret A accounts for 85% (the remaining accounts are

LDD and LEP). Banks keep 100% of all other types of regulated deposits.[6]

The share of eligible funds that have to be transferred to the CDC is set by law, and varies across banks and over time. This share used to be substantially higher for the three historical (incumbent) banks, and is required by law to converge to a single rate of 60% for all banks by 2022.[7] Table 1 summarizes the evolution of the percentages of deposits rechanneled to the CDC over time. In our empirical strategy, we use the net amount of all regulated deposits, after transfers, to measure the actual amount of deposits banks have to remunerate. By using post-transfer deposit ratios, we exploit quasi-randomness among regulated-deposit dependent banks due to the government-imposed transfer rates to the CDC. We stop the sample period before 2016 because after July 2016 banks were offered the possibility to channel all their regulated deposits to the CDC.[8]

Average funding costs and implications for banks' marginal cost of lending. As explained above, banks cannot readily adjust their exposure to regulated deposits even in the medium run due to both high demand from households and strict regulation regarding the distribution of regulated deposits and how much banks retain on their balance sheet. This implies that this source of funding is inframarginal for banks, and a change in their unit cost will only affect banks' *average* cost of funding.

In our setting, the variation in the cost of regulated deposits is set by the government rather than by the bank, and squeezes banks' net interest margin as would a fixed cost of operation or operating license fee, independent of scale, which ultimately depletes banks' expected net worth. In frameworks that relate banks' credit supply to changes in their expected net worth (e.g., Gertler and Kiyotaki, 2010; Bahaj and Malherbe, 2020), a change in the latter affects the current marginal cost of funding because lower expected net worth increases the likelihood of a bank's bankruptcy. As a result, through its effect on bank net

[6] There are also some limitations on how livret-A deposits can be used. Banks have the legal obligation to devote at least 80% of the deposits to SME lending, which motivates our sample selection in the credit-registry and firm-level data. In practice, this obligation has not been binding as the ratio of outstanding amounts of credit to SMEs to livret-A deposits has been fluctuating between 210% and 250% over the period 2010–2015.

[7] The initial target T_{bt} was 65%, and it was revised to 60% in 2013. In exchange for collecting livret-A funds, the CDC pays banks an intermediary commission proportional to the total amount of deposits.

[8] This was revoked in early 2018, and the rate of 60% has been reinforced since then.

worth, a change in a bank's inframarginal cost of funding can affect credit supply.

2.2 Data Description

Credit data. Our main data source is the French national Central Credit Register (CCR) administered by the Banque de France, whose coverage and data quality are documented in Vinas and Zapha (2026). The dataset contains monthly information on outstanding amounts of credit at the firm-branch level, granted by all credit institutions to all non-financial firms based in France, provided the total exposure (i.e., the sum of all credit of any kind and credit guarantees) of a bank to a firm exceeds €25,000. Credit is broken down by initial maturity (above and below one year). We focus on French small and medium-sized enterprises (SMEs), for which we observe the near universe; we drop those belonging to the financial sector and to public administrations, and only keep firms with standard legal forms (i.e., we drop unions, parishes, cooperatives, etc.).

We use data from 2010 to 2015 for our analysis. Our sample comprises 220 distinct banks, each of which has on average 651 branches (which can be located in the same city). For each firm, we aggregate credit across all of a given bank's branches in a given county to the bank-county level.[9] We aggregate the monthly dataset at the quarterly level to merge it with deposit data available at that frequency. The level of observation in our final dataset is the firm-bank-county-quarter level $fbct$, summarizing information on the lending relationship between firm f and bank b's branch(es) in county c in quarter t.

At the more aggregate bank-county-quarter level, we use the Cefit dataset from the Banque de France, which contains information on all outstanding amounts of credit and deposits, including loans to households and self-employed individuals that are not covered by our credit-registry data.

Deposit data. Our primary source of deposit data is regulatory data (Surfi), maintained by the ACPR. The data are available at the quarterly frequency from Q3 2010 to Q4 2015

[9] We use the definition of a French "département," which partitions the country into 100 counties. As fewer than 1% of the firms in our sample are banking with multiple branches within the same bank-county cluster, the firm-bank-county level is effectively the same as the firm-bank-branch level.

for all banks operating in France. The dataset includes deposit amounts, aggregated at the bank level b, and broken down by types of deposits (regulated vs. others) and depositors (firms, households, non-profit organizations, insurance companies and pension funds, administrations).[10]

We adjust our deposit ratios so as to take into account the net amount of eligible deposits, i.e., after rechanneling to the CDC, in the following way. Let T_{bt} be the percentage of deposits bank b has to rechannel to the CDC in year t, then: $\textit{Net eligible deposits}_{bt} = \textit{Eligible regulated deposits}_{bt} \times (1 - T_{bt})$. T_{bt} varies based on whether banks used to distribute livret-A accounts prior to the reform of 2008 (incumbent banks) or whether they were authorized to offer livret-A accounts only after 2008 (new banks).

We use the average observed percentage of funds being transferred by banks in both groups at the end of a calendar year t to define T_{bt}, i.e., we use one percentage for new banks and another one for all incumbent banks but La Banque Postale (LBP).[11] We define the regulated-deposit ratio of bank b in quarter t as:

$$\textit{Deposit ratio}_{bt} = (\textit{Non-eligible deposits}_{bt} + \textit{Net eligible deposits}_{bt}) / \textit{Total liabilities}_{bt}.$$

Firm balance-sheet data and credit ratings. Firm accounting data for SMEs come from the Fichier bancaire des entreprises (FIBEN) dataset of the Banque de France (Vinas and Zapha, 2026), and consist of firm balance sheets compiled from tax returns. The dataset includes all French firms with sales of €750,000 or more.[12]

We add firm credit-rating information for FIBEN firms using the credit ratings produced by the Banque de France. The latter assigns credit ratings to all French non-financial compa-

[10] For one bank holding company (Crédit Agricole SA), regulated, but not ordinary, deposits are reported only on a semi-consolidated basis (at a level of aggregation between the bank and the BHC level). We allocate to each bank of the respective group a fraction of the aggregate regulated-deposit amount proportional to its share out of all ordinary deposits.

[11] Given that LBP was not active in corporate lending at the beginning of the period, and could not fulfill its obligations with respect to SME lending, it was authorized to transfer all of its livret-A deposits to the CDC. We thus discard LBP from our estimations by applying a 100% transfer rate. Including it without adjusting the rate of deposits for the rechanneling scheme or including it while applying the same transfer rate as for other incumbent banks does not change the results.

[12] We drop firms with negative debt and/or negative or zero total assets. All ratios are winsorized at the 1st and 99th percentiles.

Table 2: Summary Statistics

Panel A: Main sources of variation & bank-level variables	Mean	p5	p25	Median	p75	p95	Std. dev.	*N*
Deposit ratio$_{bt}$ (Q4 2010 − Q4 2015)	0.14	0.00	0.00	0.15	0.25	0.34	0.12	3,673
Total deposit ratio$_{bt}$	0.51	0.06	0.37	0.51	0.68	0.92	0.24	3,673
Assets$_{bt}$ in billion €	32.39	0.19	1.41	8.25	16.44	116.81	122.31	3,673
Equity ratio$_{bt}$	0.04	0.00	0.01	0.02	0.04	0.12	0.07	3,673
Liquidity ratio$_{bt}$	0.01	0.00	0.00	0.01	0.01	0.04	0.05	3,673
$\frac{\text{Loans to households}}{\text{Total loans}}_{bt}$	0.22	0.00	0.00	0.00	0.48	0.71	0.28	3,673
Gap$_t$ in % (Jan 2010 − Dec 2015)	1.47	0.95	1.20	1.35	1.75	2.25	0.40	72
Panel B: Firm-bank-county-quarter level								
Credit in thousand €	397.87	28.00	54.00	119.00	287.00	1166.00	3,044.31	4,134,974
Panel C: Bank-county-quarter level								
$\frac{\text{Large firms}}{\text{Total loans}}$	0.07	0.00	0.02	0.04	0.08	0.25	0.10	28,063
$\frac{\text{Small firms}}{\text{Total loans}}$	0.09	0.00	0.04	0.07	0.11	0.23	0.08	28,063
$\frac{\text{Loans to self-employed}}{\text{Total loans}}$	0.08	0.00	0.02	0.06	0.13	0.20	0.07	28,063
$\frac{\text{High-bankruptcy industries}}{\text{Total loans}}$	0.29	0.03	0.18	0.26	0.36	0.62	0.17	27,139
$\frac{\text{Risky firms}}{\text{Rated firms}}$	0.60	0.21	0.49	0.61	0.74	0.97	0.21	26,336
$\frac{\text{MLT loans}}{\text{Total loans}}$	0.87	0.68	0.86	0.90	0.92	0.96	0.12	28,063
Panel D: ZIP-code-quarter level								
Deposit ratio$_{kt}$	0.21	0.11	0.18	0.22	0.25	0.29	0.06	664,654
Total credit in thousand €	5,353.22	61.00	294.00	834.00	2,496.00	15,827.00	59,609.23	664,654
Panel E: Firm-year level								
Deposit ratio$_{ft}$	0.12	0.00	0.03	0.13	0.21	0.29	0.10	380,657
Capital assets in million €	2.74	0.10	0.38	0.84	1.94	7.78	16.07	380,657
PP&E in million €	2.36	0.07	0.27	0.63	1.55	6.72	15.69	380,657
$\frac{\text{CapEx}}{\text{Capital assets}}$	0.23	0.00	0.03	0.09	0.24	0.87	0.46	380,657
$\frac{\text{Tangible investment}}{\text{PP\&E}}$	0.14	0.00	0.02	0.05	0.14	0.51	0.28	380,657
Employment	28.53	5.00	12.00	18.00	34.00	86.00	32.91	380,657

Notes: In Panel A, *Deposit ratio*$_{bt}$ is the ratio of regulated deposits over total liabilities of bank b in quarter t; *Total deposit ratio*$_{bt}$ is the ratio of all deposits over total liabilities of bank b in quarter t; *Assets*$_{bt}$ denotes total assets of bank b in quarter t; *Equity ratio*$_{bt}$ is the ratio of equity over total assets of bank b in quarter t; *Liquidity ratio*$_{bt}$ is the ratio of cash and central-bank reserves (i.e., liquid assets) over total assets of bank b in quarter t; $\frac{\textit{Loans to households}}{\textit{Total loans}}_{bt}$ is the fraction of household lending out of total lending by bank b in quarter t; and *Gap*$_t$ is the difference between the rate on regulated deposits (livret A) and the ECB's deposit facility rate in month t. The summary statistics in Panels B, C, D, and E correspond to Table 5, Table 8, Table 9, and Table 10, respectively, and the sample period is Q4 2010 to Q4 2015 (Table 5, Table 8, and Table 9) and 2010 to 2015 (annual data, Table 10).

nies with at least three consecutive years of accounting data. The main use of the ratings is to determine the eligibility of bank loans to rated firms as collateral for Eurosystem funding (see Cahn, Duquerroy, and Mullins, 2019, for more details), and provide an assessment of firms' ability to meet their financial commitments over a three-year horizon.

Summary statistics. Table 2 presents summary statistics for all relevant samples and variables. In Panel A, we zoom in on our main sources of variation, namely bank-level variables, such as banks' regulated-deposit ratios, and the gap between the rate on regulated deposits (livret A) and the ECB's deposit facility rate. Regulated deposits account for almost one-third of total deposits and, thus, constitute an important source of retail funding. Gap_t ranges from 0.95 to 2.25 percentage points between the 5th and 95th percentiles, with a standard deviation of 0.4 percentage points, and we use its level at the end of each quarter in our analysis.

In Panel B, we move to the firm-bank-county-quarter level, the level of observation for all credit-registry-based regressions. On this basis, we aggregate data up to the ZIP-code-quarter level in Panel D. The aggregation at the bank-county-quarter level in Panel C is based on the Cefit dataset.[13] Finally, in Panel E, we include summary statistics for all outcome variables at the firm-year level for firms with balance-sheet data.

We also present summary statistics separately for banks with regulated-deposit ratios in the top and bottom half of the distribution in Table 3. Banks with higher regulated-deposit ratios are smaller in terms of assets, generally more dependent on deposits, and source their deposits primarily from households rather than corporations, whereas the opposite holds for banks with lower regulated-deposit ratios. We formally test the correlation between regulated-deposit ratios and other bank covariates in Figure B.2 of the Online Appendix, and discuss the implications for the empirical identification in Section 2.3.

External validity. Our setting is representative of bank-based advanced economies. Among the 22 advanced economies in the World Bank's Global Financial Development Database,

[13] In Panel C, firms' average ratings, which are used to identify risky firms, are calculated from rating data merged with the credit registry.

Table 3: High- vs. Low-regulated-deposit Banks

Banks with regulated-deposit ratios in the top half	Mean	p5	p25	Median	p75	p95	Std. dev.	N
Total deposit ratio$_{bt}$	0.59	0.38	0.48	0.56	0.68	0.93	0.15	1,836
$\frac{\text{Household deposits}}{\text{Total deposits}}$	0.56	0.34	0.42	0.52	0.72	0.85	0.17	1,836
$\frac{\text{Corporate deposits}}{\text{Total deposits}}$	0.34	0.10	0.22	0.38	0.44	0.50	0.13	1,836
Total loans in billion €	10.88	0.34	5.40	8.17	12.00	26.22	14.30	1,836
Corporate loans in billion €	2.65	0.10	1.23	2.05	3.07	6.31	3.34	1,836
Loans to households in billion €	3.02	0.00	0.00	0.00	3.50	9.93	8.04	1,836
Loans to self-employed in billion €	1.03	0.02	0.34	0.75	1.37	2.28	1.42	1,836
$\frac{\text{MLT loans}}{\text{Total loan portfolio}}$	0.90	0.84	0.89	0.91	0.92	0.94	0.04	1,836
$\frac{\text{MLT corporate loans}}{\text{Corporate loan portfolio}}$	0.57	0.40	0.52	0.58	0.65	0.71	0.11	1,836
Equity ratio$_{bt}$	0.02	0.00	0.01	0.02	0.03	0.07	0.02	1,836
Assets$_{bt}$ in billion €	18.25	0.65	7.97	12.50	18.62	53.96	28.40	1,836
Banks with regulated-deposit ratios in the bottom half								
Total deposit ratio$_{bt}$	0.44	0.01	0.19	0.44	0.65	0.92	0.28	1,837
$\frac{\text{Household deposits}}{\text{Total deposits}}$	0.32	0.00	0.02	0.35	0.52	0.83	0.27	1,837
$\frac{\text{Corporate deposits}}{\text{Total deposits}}$	0.58	0.11	0.36	0.55	0.89	1.00	0.29	1,837
Total loans in billion €	7.94	0.07	0.40	1.29	4.39	29.66	24.52	1,837
Corporate loans in billion €	3.11	0.02	0.11	0.54	1.66	13.13	8.71	1,837
Loans to households in billion €	2.70	0.00	0.00	0.00	1.30	9.83	9.11	1,837
Loans to self-employed in billion €	0.39	0.00	0.00	0.00	0.08	2.25	1.33	1,837
$\frac{\text{MLT loans}}{\text{Total loan portfolio}}$	0.63	0.03	0.40	0.76	0.89	0.97	0.31	1,837
$\frac{\text{MLT corporate loans}}{\text{Corporate loan portfolio}}$	0.51	0.00	0.36	0.50	0.66	1.00	0.28	1,837
Equity ratio$_{bt}$	0.06	0.00	0.01	0.03	0.07	0.19	0.09	1,837
Assets$_{bt}$ in billion €	46.52	0.14	0.72	2.42	9.64	303.15	169.45	1,837

Notes: All variables are measured at the bank-quarter level bt. Summary statistics in the top (bottom) panel are for banks with ratios of regulated deposits over total liabilities in the top (bottom) half of the distribution. *Total deposit ratio*$_{bt}$ is the ratio of all deposits over total liabilities of bank b in quarter t. Summary statistics on banks' lending activity correspond to the respective descriptions in Table 8, with the exception of $\frac{MLT\ corporate\ loans_{bt}}{Corporate\ loan\ portfolio_{bt}}$, which is the ratio of bank b's corporate loans with a maturity of more than one year over its total corporate-loan exposure (based on the data in Table 5). *Equity ratio*$_{bt}$ and *Assets*$_{bt}$ are, respectively, the ratio of equity over total assets and total assets of bank b in quarter t.

France is close to the median in bank concentration, private bank credit, and bank deposits. French firms also use banks at least as much as firms elsewhere (Figure D.4). Appendix D reports the full comparison.

Beyond these similarities, while the exact setting is specific to France, what we estimate conceptually is the elasticity of credit supply to an inframarginal shock to bank profits, since a higher livret-A rate raises the cost of a deposit base that banks cannot adjust, while the marginal cost of funding a new loan is unchanged. We would therefore expect a similar elasticity in other settings where banks face inframarginal shocks to their profits or to the value of their deposit franchise. Because the shock raises the cost of maintaining a deposit franchise, our estimates also inform the debate on whether the franchise hedges banks against interest-rate risk (Drechsler, Savov, and Schnabl, 2021). This cost passes through to credit supply, so the operating cost of the franchise affects lending as well as margins, in line with the concerns raised by Begenau and Stafford (2022) and DeMarzo, Krishnamurthy, and Nagel (2024), and the fragility documented by Jiang, Matvos, Piskorski, and Seru (2024).

2.3 Identification

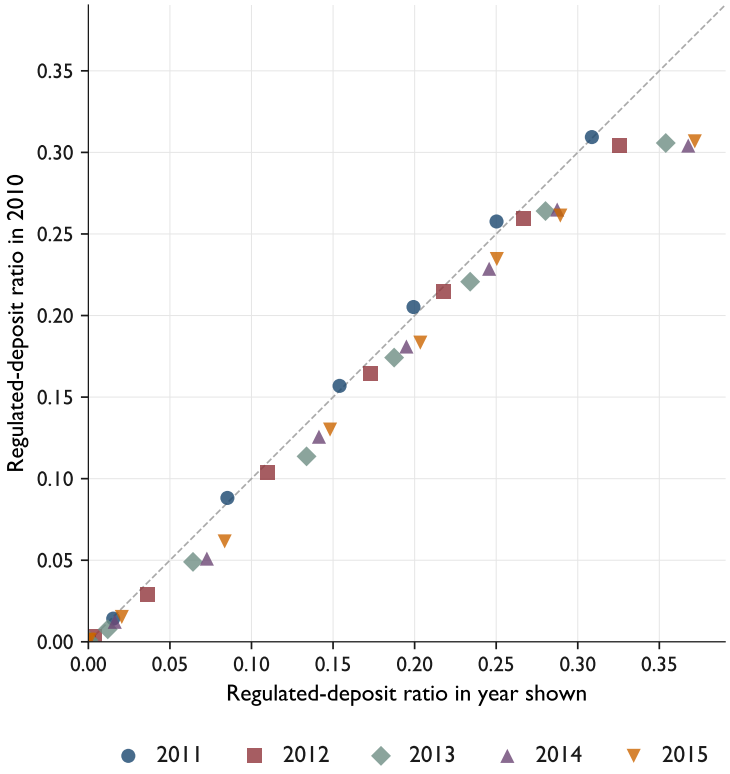
We use the following specification to estimate how banks' funding costs affect their lending:

$$\begin{aligned} \ln(Credit)_{fbct} &= \beta_1 Deposit\ ratio_{bt-1} \times Gap_t + \beta_2 Deposit\ ratio_{bt-1} \\ &\quad + \mu_{fbc} + \theta_{ft} + \psi_{ct} + \epsilon_{fbct}, \end{aligned} \tag{1}$$

where $Credit_{fbct}$ measures the euro amount of debt outstanding between firm f and bank b's branch(es) in county c in quarter t, $Deposit\ ratio_{bt-1}$ is the ratio of regulated deposits over total liabilities of bank b in quarter $t-1$, which is assigned to all branches of bank b, and Gap_t is the difference between the rate on regulated deposits (livret A) and the ECB's deposit facility rate (in %) at the end of quarter t. μ_{fbc}, θ_{ft}, and ψ_{ct} denote firm-bank-county, firm-quarter, and bank b's county-quarter fixed effects, respectively. We cluster standard errors at the bank level, which corresponds to our level of identifying variation.

Our coefficient of interest, β_1, reflects the elasticity of banks' credit supply with respect to

Figure 2: **Variation of Regulated-deposit Ratios within Banks over Time.** For each year from 2011 to 2015, banks are sorted into 20 equal-sized groups by their regulated-deposit ratio, *Deposit ratio*$_{bt}$, measured in the fourth quarter of that year. Each marker plots the group's mean ratio in that year (x-axis) against the same banks' mean ratio in the fourth quarter of 2010 (y-axis). Banks with a ratio of zero form a single group, so each year has fewer than 20 markers; every group contains at least five banks. The dashed line is the 45-degree line.



their funding costs, measured in our setting by the cost of regulated deposits relative to other sources of funding (e.g., ordinary deposits or interbank funding). Under the assumption that otherwise-funded banks experience perfect pass-through of the ECB's deposit facility (DF) rate to their funding costs, *Deposit ratio*$_{bt-1} \times$ *Gap*$_t$ is the difference in average funding costs between banks more dependent on regulated deposits and all other banks, including those that rely exclusively on other sources of funding.

Figure 2 plots the persistence in the share of regulated deposits over different horizons (from 2010 to 2011 up to 2015). The markers lie close to the 45-degree line, with deviations that can be explained by the exogenously imposed time variation in transfer rates to the CDC (see Table 1). The persistence of banks' regulated-deposit ratio implies that banks cannot adjust their exposure to regulated deposits even in the medium run, which has two consequences. First, β_1 is unlikely to reflect differences in their ability to substitute across

marginal sources of funding. Second, a change in the cost of regulated deposits is equivalent to a shift in banks' operating costs including those to maintain a stable deposit base, i.e., the deposit franchise.

Our regression specification with multiple high-dimensional fixed effects addresses several potential sources of endogeneity. μ_{fbc} are borrower-by-bank county (i.e., comprising all branches of a given bank in a given county) fixed effects that remove time-invariant unobserved heterogeneity across borrower-lender pairs. This also implies that our treatment effect is estimated only for the intensive margin, within an existing borrower-lender pair, and does not depend on the creation/destruction of new bank-firm relationships.

We control for time-varying unobserved heterogeneity across firms that might affect their credit *demand* by including borrower-by-quarter fixed effects θ_{ft}. The cost of doing so is that our coefficient of interest is only identified for firms borrowing from multiple lenders, as otherwise the time-varying bank-level shock would be perfectly collinear with the firm-by-quarter fixed effects.

Because borrowers are not necessarily located in the same county as the bank branches from which they obtain loans, we can also include bank county-by-quarter fixed effects ψ_{ct}.[14] This set of fixed effects controls for time-varying unobserved differences across counties where the credit-granting branches of bank b are based. Therefore, β_1 is estimated by comparing different banks (and their branches) in the same county lending to the same firm over time.[15]

Threat to identification. Our estimation does not require banks' regulated-deposit ratios to be as good as randomly assigned. Table 3 shows that they are not, since banks with high ratios are smaller and fund themselves mostly with household deposits. Firm-bank-county fixed effects absorb any time-invariant consequence of these differences, and $\textit{Deposit ratio}_{bt-1}$ enters on its own, so a bank's exposure can be correlated with the level of its lending without biasing β_1. What we require is that, absent changes in the relative cost of regulated deposits,

[14] Within the subset of firms borrowing from multiple banks, 38% borrow from at least one bank located in a different county.

[15] What is more, if firms borrowing from multiple banks across different counties share the same motivation for additionally sourcing credit from a bank branch in another county, bank county-by-quarter fixed effects also capture location-specific credit demand.

banks with higher deposit ratios would not have changed their lending differently in the quarters in which Gap_t rises and falls. A confounder would therefore have to affect banks in proportion to their deposit ratio and feature the same fluctuations as Gap_t.

We argue that such a confounder is hard to find since Gap_t is set by the government, which often departs from the formula (Section 2.1.1), and its correlation with the ECB's deposit facility rate between 2010 and 2014 is -0.01 (Figure 1). It also moves in both directions, ranging from 150 to 225 basis points between Q4 2011 and Q2 2013 and dropping to around 100 basis points thereafter. A shock that follows the monetary-policy cycle, or that trends differently for banks with high deposit ratios, would not reproduce this pattern.

The remaining threat is a time-varying bank-level shock that co-moves with Gap_t and loads on banks' deposit ratios. While it is impossible to fully solve this problem since this is our level of identifying variation, we partially address this issue in three ways.

First, we estimate eq. (1) with banking group (BHC)-by-quarter fixed effects,[16] which restrict the variation to banks belonging to the *same* banking group and, thus, control for time-varying unobserved differences at this more aggregate level that may affect credit supply (e.g., differences in banking-group business models or broader wholesale funding shocks).

Second, we test for the correlation between our exposure measure and other bank characteristics and report this balance test in Figure B.2 of the Online Appendix. We regress banks' initial regulated-deposit ratio, one characteristic at a time, on ten characteristics that cover size (log total assets), capitalization (equity over assets), liquidity (the liquidity ratio), profitability (profits over revenues, return on assets, and net interest margin), deposit funding (total deposits over liabilities and the volatility of deposit growth), net interbank lending, and business model (household loans over total loans). Regulated-deposit dependent banks are unconditionally different, as 9 of the 10 characteristics are significant at the 5% level when we include only quarter fixed effects. However, they become similar along most dimensions once we control for the mechanical correlation between regulated-deposit dependence and reliance on deposits, through banks' total-deposit share interacted with quarter fixed effects. The only two characteristics that remain statistically different from zero at the 5%

[16] We have 69 banking groups in our sample.

level are the liquidity ratio and profits over revenues.[17] In columns 3 to 5 of Table A.7 of the Online Appendix, we use a more flexible specification that interacts each control with quarter fixed effects.

Table 4: Average Effect of Funding Costs on Interest Paid and Net Income: Difference-in-Differences

Dependent variable	Average interest paid (pp)				ln(Net income)			
Treatment	Dep. ratio Q3 2010		Top 50%		Dep. ratio Q3 2010		Top 50%	
Bank controls	No	Yes	No	Yes	No	Yes	No	Yes
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treatment × High-gap period	1.090***	1.108***	0.123**	0.122**	-1.866***	-1.736***	-0.244*	-0.218*
	(0.213)	(0.237)	(0.049)	(0.051)	(0.648)	(0.563)	(0.131)	(0.114)
Fixed Effects								
Bank	✓	✓	✓	✓	✓	✓	✓	✓
Banking group × year	✓	✓	✓	✓	✓	✓	✓	✓
Observations	420	420	420	420	411	411	411	411

Notes: The level of observation is at the bank-year level bt. The sample period is 2011 to 2015. In columns 1–4, the dependent variable is bank b's average interest paid on deposits in year t, annual deposit interest expense divided by the average deposit balance, in percentage points and winsorized at the 1st and 99th percentiles. In columns 5–8, it is the natural logarithm of bank b's income-statement net income in year t. $Treatment_b$ is a time-invariant characteristic at the bank level b: the ratio of regulated deposits over total liabilities of bank b in Q3 2010 (columns 1, 2, 5 and 6), or an indicator for whether that ratio is in the top half of the bank-level distribution (columns 3, 4, 7 and 8). $High\text{-}gap\ period_t$ is a dummy variable for 2011 and 2012, which are characterized by high values of Gap_t, the difference between the rate on regulated deposits (livret A) and the ECB's deposit facility rate. Whenever indicated, controls include bank b's liquidity ratio, natural logarithm of total assets, and share of non-performing assets, all measured in year $t-1$. Samples differ across the two outcomes because they exclude observations with missing inputs and fixed-effect singletons; columns 5–8 also require positive net income. Robust standard errors (clustered at the bank level) are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Third, we allow the effect of Gap_t to differ between quarters in which the government deviates from the livret-A formula and quarters in which it follows the formula.

3 Results

3.1 Banks' Interest Paid and Profits

We first test whether shifts in Gap_t change the interest that regulated-deposit dependent banks pay on their deposits, and how this affects their profits. We estimate a difference-in-differences specification at the bank-year level bt, at which we observe interest paid and net income. Treatment is a bank's regulated-deposit ratio in Q3 2010, measured either continuously or as an indicator for the top half of the distribution, and the high-gap period comprises 2011 and 2012, when Gap_t is high, against 2013 to 2015. All specifications include

[17] We control for both in columns 5 to 7 of Table 5, where each enters in levels and interacted with Gap_t.

Table 5: Average Effect of Funding Costs on Credit Supply

Dependent variable	ln(Credit)									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Deposit ratio	0.106	0.137	0.113	0.293*	0.140	0.123	0.301*	-0.029	0.137	0.115
	(0.094)	(0.137)	(0.099)	(0.177)	(0.137)	(0.137)	(0.175)	(0.134)	(0.137)	(0.125)
Deposit ratio × Gap	-0.084***	-0.160***	-0.162***	-0.144***	-0.164***	-0.145***	-0.139***			
	(0.026)	(0.043)	(0.046)	(0.049)	(0.045)	(0.044)	(0.047)			
Bank size × Gap			0.001							
			(0.002)							
Total deposits / liabilities × Gap				0.024			0.052***			
				(0.022)			(0.020)			
Liquidity ratio × Gap					-0.216		-0.131			
					(0.172)		(0.164)			
Profits / revenues$_{t-1}$ × Gap						-0.105**	-0.179***			
						(0.050)	(0.060)			
Deposit ratio × Gap in top tercile								-0.124***		
								(0.047)		
Deposit ratio × Gap in 2nd tercile								-0.014		
								(0.032)		
Deposit ratio × $\widetilde{\text{Gap}}$									-0.062***	
									(0.017)	
Deposit ratio × $\widetilde{\text{Gap}}$ ×D										-0.058***
										(0.015)
Deposit ratio × $\widetilde{\text{Gap}}$ ×$(1-D)$										-0.066***
										(0.020)
Fixed Effects										
Firm-bank-county	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Firm-quarter	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
County-quarter	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Banking group × quarter		✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	4,495,354	4,495,354	4,495,354	4,495,354	4,495,354	4,495,035	4,495,035	4,495,354	4,495,354	4,495,354

Notes: The level of observation is credit to firm f by bank b's branch(es) in county c in quarter t. The sample period is Q4 2010 to Q4 2015. The dependent variable is the natural logarithm of the euro amount of debt outstanding between firm f and bank b's branch(es) in county c in quarter t. *Deposit ratio*$_{bt-1}$ is the ratio of regulated deposits over total liabilities of bank b in quarter $t-1$. Gap_t is the difference between the rate on regulated deposits (livret A) and the ECB's deposit facility rate (in %) at the end of quarter t. Each bank control enters both in levels and interacted with Gap_t; only the interactions are reported. Column 1 omits banking group × quarter fixed effects. In column 3, *Bank size*$_{bt-1}$ is the natural logarithm of total assets of bank b in quarter $t-1$. In columns 4–7, *Total deposits / liabilities*$_{bt-1}$ is total deposits over total liabilities, and profits over revenues are the prior-year annual value, held constant within the year; liquidity and profits over revenues are winsorized at the 1st and 99th percentiles, and columns 6 and 7 drop 189 observations without prior-year profits over revenues and 130 fixed-effect singletons. In column 8, *Gap in top* (2nd) *tercile*$_t$ is a dummy variable for whether Gap_t ranges in the top (middle) tercile of its distribution. In columns 9 and 10, $\widetilde{\text{Gap}}_t$ is Gap_t divided by its standard deviation across the 21 sample quarters (0.385); column 9 is therefore column 2 rescaled. In column 10, D_t equals one in quarters in which the livret A rate deviates from its mechanical formula (quarterly average of actual minus formula rate nonzero); both interactions enter jointly. Robust standard errors (clustered at the bank level) are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

bank as well as banking group-by-year fixed effects. Columns 1 to 4 of Table 4 show that treated banks pay more on their deposits in the high-gap years: banks in the top half of the regulated-deposit distribution pay 0.123 percentage points more than the other banks (column 3, standard error 0.049). The higher cost reduces banks' net income by ($1-e^{-0.244}=$) 21.7% for banks in the top half (column 7). Columns 2, 4, 6, and 8 add lagged bank-level controls to rule out that our identifying variation reflects a correlation with time-varying bank characteristics. The point estimates are barely affected: the coefficient on net income moves from -1.866 to -1.736 with the continuous treatment, and from -0.244 to -0.218 with the top-half indicator.

An increase in Gap_t therefore raises the interest that regulated-deposit dependent banks pay, and lowers their profits. We next study how these banks adjust their credit supply in response.

3.2 Average Effect on Credit Supply

In the first column of Table 5, we estimate eq. (1), using as $Deposit\ ratio_{bt-1}$ the ratio of regulated deposits over total liabilities in quarter $t-1$. We find that regulated-deposit dependent banks reduce their lending when the interest they have to pay on these deposits increases. This estimate becomes larger after the inclusion of BHC-quarter fixed effects in column 2 (our preferred specification), which suggests imperfect internal capital markets within banking groups.[18] As $Deposit\ ratio_{bt-1} \times Gap_t$ measures the difference in funding costs incurred by any bank with non-zero regulated deposits vs. banks whose cost of funding is aligned with the monetary-policy rate, our estimate in column 2 implies that banks contract their lending by 16.0% if their average funding costs are one percentage point higher.

To gauge the magnitude in the absence of comparable estimates in the literature (mainly due to the lack of high-frequency funding-cost shocks that also vary in size), one can make use of the fact that French banks have virtually no control over their exposure to regulated deposits and the cost thereof. As such, $Deposit\ ratio_{bt-1} \times Gap_t$ is equivalent to a fixed

[18] If banking groups were able to reallocate resources well across their different banks, we should find a smaller, i.e., less negative, point estimate in column 2, as the reallocation would allow banks belonging to the same group to immunize themselves against any bank-level shocks that could affect their lending.

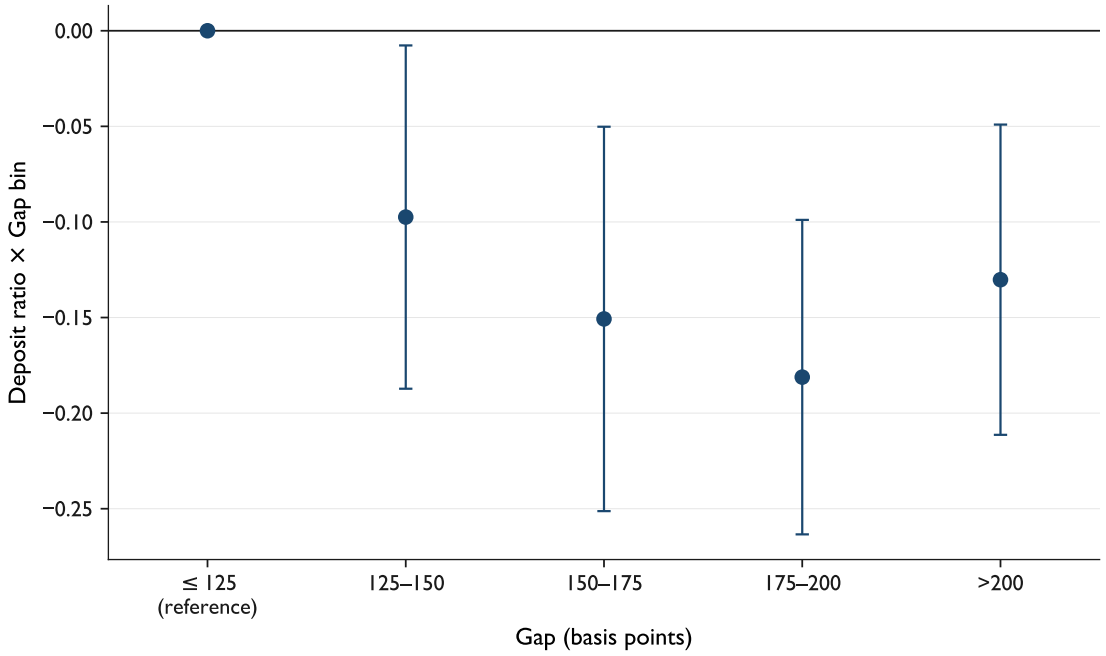
cost of operation that cuts into bank profits and can, thus, be compared to changes in bank equity. As the average bank holds 14% of its liabilities in regulated deposits and 4% of its assets in equity (see Panel A of Table 2), for every percentage-point change in Gap_t, equity fluctuates by 14% $\times$ 96 (liabilities) divided by 4 (equity) percent, which equals 3.36%. The lending response in column 2 then implies an elasticity of $((14\% \times 16.0\%)/3.36\% =)$ 0.67. In terms of one standard deviation of Gap_t (38.5 basis points), equity falls by $(0.385 \times 3.36\% =)$ 1.29% for the average bank, and lending falls by $(14\% \times 0.062 =)$ 0.87% (column 9).

In column 4, we include as a control variable *Total deposits / liabilities*$_{bt-1}$, the ratio of all deposits, including regulated deposits, over total liabilities of bank b in quarter $t-1$, and its interaction with Gap_t. This allows us to hold constant any shared characteristics of banks relying more on deposits (regulated or not) that could govern credit-supply responses to fluctuations in Gap_t. The effect of *Deposit ratio*$_{bt-1} \times Gap_t$ remains quantitatively similar (-0.144, compared with -0.160 in column 2), while the point estimate for *Total deposits / liabilities*$_{bt-1} \times Gap_t$ is 0.024 with a standard error of 0.022.

In columns 3 and 5 to 7, we control directly for the bank characteristics on which regulated-deposit dependent banks differ in the balance test of Figure B.2 once we control for the share of deposits in liabilities. Because Gap_t is measured relative to the monetary-policy rate, we focus on characteristics that govern the transmission of monetary policy to credit supply through bank net worth. Size (Kashyap and Stein, 1995) is the largest unconditional difference: with quarter fixed effects only, a one-standard-deviation increase in log total assets is associated with a regulated-deposit ratio that is 0.67 standard deviations lower. This difference is no longer significant once we control for banks' total-deposit share, whereas liquidity and profitability remain different at the 5% level. In column 3, we add log total assets and its interaction with Gap_t. In columns 5 and 6, we add banks' liquidity ratio and their prior-year profits over revenues in the same way, and column 7 controls jointly for total deposits, liquidity, and profitability. Across these four columns, the coefficient on *Deposit ratio*$_{bt-1} \times Gap_t$ ranges from -0.139 to -0.164, against -0.160 in column 2.[19]

[19] In Table A.7 of the Online Appendix, we also control nonparametrically for total deposits, liquidity, and profitability by interacting each with quarter fixed effects, and find similar results: the coefficient ranges from -0.128 to -0.166.

Figure 3: **Lending Response by Level of the Funding-cost Gap.** This figure plots the coefficients on *Deposit ratio*$_{bt-1}$ interacted with indicators for bins of Gap_t from the specification in column 2 of Table 5, with bins of 25 basis points above 125 basis points. Quarters in which Gap_t is at most 125 basis points are the reference. Bars show 95% confidence intervals based on standard errors clustered at the bank level.



In column 8, we estimate the effect of a change in the relative cost of regulated deposits nonparametrically by replacing Gap_t with two indicator variables that equal one if Gap_t belongs to the top or middle tercile of its distribution, respectively. The top tercile comprises all observations with a value of Gap_t of at least 150 basis points, and the middle tercile comprises all observations with a value of Gap_t of at least 120 (but less than 150) basis points. Therefore, the coefficient on *Deposit ratio*$_{bt-1}$ now captures the effect for regulated-deposit dependent banks when Gap_t is less than 120 basis points.

Nonlinearity. The effect of funding costs on credit supply is highly nonlinear. Relative to the bottom tercile, in which Gap_t is below 120 basis points, the coefficient for the middle tercile is −0.014 (standard error 0.032), and only the top tercile shows a significant contraction (−0.124, standard error 0.047). Figure 3 uses finer bins of 25 basis points, so its reference group comprises the quarters in which Gap_t is at most 125 basis points. The response is already significant at the 5% level between 125 and 150 basis points (−0.097, standard error 0.046), grows to −0.151 between 150 and 175 basis points and to −0.181 between 175 and

Table 6: Average Effect of Funding Costs on Credit Supply: Difference-in-Differences

Treatment definition	ln(Credit) Dep. ratio Q3 2010 (1)	ln(Credit) Top 50% (2)	ln(Credit) Top 25% (3)	ln(Credit) Incumbent banks (4)
Treatment × High-gap period	-0.210***	-0.019**	-0.030***	-0.047***
	(0.045)	(0.008)	(0.008)	(0.008)
Fixed Effects				
Firm-bank-county	✓	✓	✓	✓
Firm-quarter	✓	✓	✓	✓
County-quarter	✓	✓	✓	✓
BHC-quarter	✓	✓	✓	✓
Observations	3,681,539	3,681,539	3,681,539	3,681,539

Notes: The level of observation is credit to firm f by bank b's branch(es) in county c in quarter t. The sample period is Q4 2011 to Q4 2015. The dependent variable is the natural logarithm of the euro amount of debt outstanding between firm f and bank b's branch(es) in county c in quarter t. $Treatment_b$ is a time-invariant characteristic at the bank level b. In column 1, it is equal to the ratio of regulated deposits over total liabilities of bank b in Q3 2010. In columns 2 and 3, it is defined as an indicator variable for whether bank b's regulated-deposit ratio in Q3 2010 is, respectively, in the top half or quartile of the bank-level distribution. In column 4, $Treatment_b$ is a dummy variable for whether bank b is one of the "incumbent banks," i.e., Caisses d'Epargne et de Prévoyance or Crédit Mutuel. *High-gap period*$_t$ is a dummy variable for the period from Q4 2011 up until (and including) Q2 2013, which is characterized by high values of Gap_t, the difference between the rate on regulated deposits (livret A) and the ECB's deposit facility rate. Robust standard errors (clustered at the bank level) are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The R-squared is 0.95 in all columns.

200 basis points, and is −0.130 in the two quarters in which Gap_t exceeds 200 basis points.

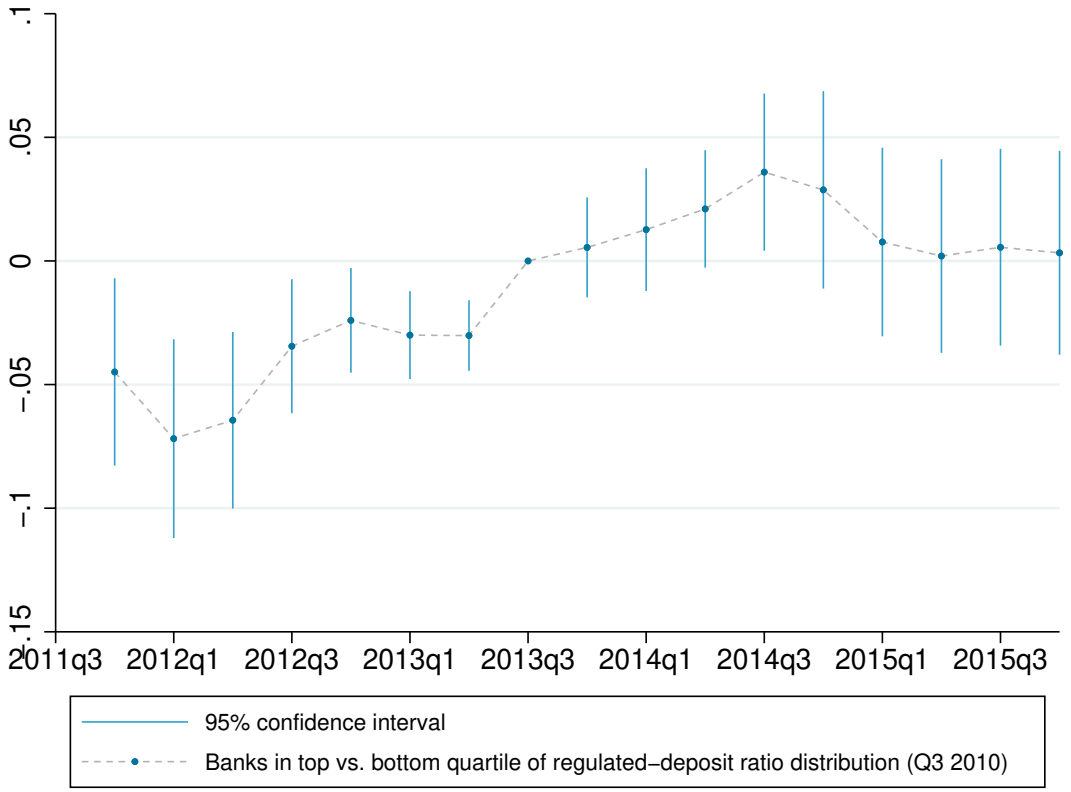
Column 9 expresses the baseline estimate in standard-deviation units by dividing Gap_t by its standard deviation across the 21 sample quarters (38.5 basis points). The coefficient of −0.062 implies that a one-standard-deviation increase in Gap_t reduces lending by (0.14 × 6.2% =) 0.87% for a bank with the average regulated-deposit ratio of 14%.

Formula and political deviations. Column 10 shows that the lending response is similar in quarters in which the government deviates from the formula and in quarters in which it follows it. The livret-A formula follows money-market rates and inflation, which could be correlated with the economic conditions that drive bank lending, whereas the deviations from it described in Section 2.1.1 are decided by the government on grounds that concern household savers rather than bank lending. We interact *Deposit ratio*$_{bt-1}$ × $\widetilde{\text{Gap}}_t$ with an indicator D_t for quarters in which the livret-A rate deviates from its formula and with its complement. The two coefficients are −0.058 (standard error 0.015) in quarters with a deviation and −0.066 (standard error 0.020) in the remaining quarters, both close to the −0.062 of column 9.

Figure 4: **Low vs. High Funding-cost Gap and Lending by Regulated-deposit Dependent Banks.** This figure plots β_k over time from estimating the following regression specification from Q4 2011 to Q4 2015 at the quarterly frequency:

$$\ln(Credit)_{fbct} = \sum_{k=1,k\neq 8}^{17} \beta_k High\ deposits_b \times D_t^k + \mu_{fbc} + \theta_{ft} + \psi_{ct} + \xi_{j(b)t} + \epsilon_{fbct},$$

where $High\ deposits_b$ equals one when bank b's regulated-deposit ratio is in the top quartile of the distribution in Q3 2010, and zero if it is in the bottom quartile, and D_t^k is an indicator variable for the kth quarter-year starting in Q4 2011 ($k = 1$), with Q3 2013 (the beginning of the low Gap_t period) being the omitted category ($k = 8$). In addition, $\xi_{j(b)t}$ denotes banking group j (of bank b) by quarter fixed effects.



Because the bank-level regulated-deposit ratio is stable over time (see Figure 2), we can replace our bank-level exposure measure by a pre-determined and time-invariant regulated-deposit ratio. In Table A.1 and Table A.2 of the Online Appendix, we use the ratio of regulated (and total) deposits over the total liabilities of bank b in Q3 2010 and Q4 2010, respectively. The estimates are similar across all specifications.

Using a pre-determined time-invariant exposure variable also allows us to estimate a difference-in-differences specification with a pre- and a post-period. For this purpose, we zoom in on the sample from Q4 2011 until Q4 2015, during the first half of which (up

until Q2 2013) Gap_t is high and ranges from 150 to 225 basis points. This is precisely the range that we have found to mark the nonlinear credit-supply response of regulated-deposit dependent banks (in column 8 of Table 5). Gap_t drops sharply to around 100 basis points thereafter (see Figure 1).

In column 1 of Table 6, we first use the time-invariant regulated-deposit ratio in Q3 2010 (as in Table A.1) and interact it with *High-gap period*$_t$, an indicator variable for the period from Q4 2011 to Q2 2013. In line with the fact that this time period is characterized by high values of Gap_t that exceed those during the remaining sample period by over one percentage point, the difference-in-differences estimate is slightly larger than our baseline estimate in column 2 of Table 5. In columns 2 and 3 of Table 6, we replace the treatment-exposure variable by indicator variables for whether bank b's regulated-deposit ratio in Q3 2010 is, respectively, in the top half or quartile of the bank-level distribution. Compared with all remaining banks, those with a regulated-deposit ratio in the top quartile of the distribution contract their lending by ~~2.9~~3.0% in the high-gap period as opposed to the low-gap period. Finally, in column 4, we obtain an even larger estimate when using the incumbent banks that offered regulated deposits well before the 2008 reform and, as such, have significantly higher regulated-deposit ratios to start with.

Figure 4 plots the event-study version of Table 6: the estimated difference in credit supply for banks in the top vs. bottom quartile of the (time-invariant) ratio of regulated deposits over total liabilities (in Q3 2010) over time. Banks dependent on regulated deposits lend less during the high-gap period until Q2 2013. Thereafter, the funding-cost gap drops markedly and then stabilizes. The credit-supply difference between highly and weakly regulated-deposit dependent banks follows a similar pattern: it stabilizes and remains indistinguishable from zero in the period starting in Q4 2013 relative to the reference quarter Q3 2013.

Role of bank heterogeneity. In Table 7, we explore the heterogeneity across banks in their credit-supply response by modifying the regression specification from column 2 of Table 5 to include interactions with different bank characteristics.

We first consider banks' capitalization, as measured by their (time-invariant) equity-to-

Table 7: The Effect of Funding Costs on Credit Supply across Bank Characteristics

Bank characteristic	ln(Credit) Equity ratio (1)	ln(Credit) Low equity (2)	ln(Credit) Liquidity ratio (3)	ln(Credit) High liquidity (4)	ln(Credit) NPL share (5)	ln(Credit) High NPL (6)
Deposit ratio × Gap × Bank characteristic	3.635**	-0.119	16.552***	0.145*	2.244	0.212**
	(1.637)	(0.080)	(5.204)	(0.087)	(1.497)	(0.095)
Deposit ratio × Gap	-0.262***	-0.130***	-0.274***	-0.196***	-0.219***	-0.194***
	(0.062)	(0.049)	(0.058)	(0.055)	(0.075)	(0.050)
Deposit ratio × Bank characteristic	-4.694	-0.074	-25.865**	-0.363	-5.151	-0.483***
	(4.857)	(0.201)	(12.441)	(0.256)	(3.206)	(0.171)
Deposit ratio	0.274	0.198	0.339	0.253	0.278	0.197
	(0.166)	(0.173)	(0.211)	(0.173)	(0.174)	(0.135)
Bank characteristic × Gap	-0.140	0.002	-1.766**	-0.019	-0.062	-0.026*
	(0.216)	(0.017)	(0.684)	(0.016)	(0.254)	(0.016)
Bank characteristic					0.317	0.062**
					(0.640)	(0.030)
Fixed Effects						
Firm-bank-county	✓	✓	✓	✓	✓	✓
Firm-quarter	✓	✓	✓	✓	✓	✓
County-quarter	✓	✓	✓	✓	✓	✓
BHC-quarter	✓	✓	✓	✓	✓	✓
Observations	4,495,354	4,495,354	4,495,354	4,495,354	4,495,354	4,495,354

Notes: The level of observation is credit to firm f by bank b's branch(es) in county c in quarter t. The sample period is Q4 2010 to Q4 2015. The dependent variable is the natural logarithm of the euro amount of debt outstanding between firm f and bank b's branch(es) in county c in quarter t. *Deposit ratio*$_{bt-1}$ is the ratio of regulated deposits over total liabilities of bank b in quarter $t-1$. Gap_t is the difference between the rate on regulated deposits (livret A) and the ECB's deposit facility rate (in %) at the end of quarter t. In the first four columns, *Bank characteristic*$_b$ is a time-invariant bank-level characteristic, namely bank b's continuous ratio of equity over total assets (column 1), an indicator for whether its equity-to-assets ratio is in the bottom tercile of the bank-level distribution (column 2), the continuous ratio of bank b's cash and central-bank reserves (i.e., liquid assets) over total assets (column 3), and an indicator for whether its ratio of cash and central-bank reserves over total assets is in the top tercile of the bank-level distribution (column 4), all measured at the beginning of the sample period (Q3 2010). In columns 5 and 6, *Bank characteristic*$_{bt-1}$ is based on bank b's share of non-performing loans (NPLs) out of total loans, and the respective variable in column 6 is an indicator for whether its share of NPLs out of total loans is in the top tercile of the bank-level distribution, in quarter $t-1$. Robust standard errors (clustered at the bank level) are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The R-squared is 0.94 in all columns.

assets ratio at the beginning of our sample. In column 1, higher funding costs depress bank lending less for strongly capitalized banks. In column 2, ~~we show there is a distinct negative effect on credit supply by low-equity banks~~the coefficient for low-equity banks, characterized as banks with equity-to-assets ratios in the bottom tercile of the distribution, is −0.119. These estimates lend support to the idea that banks' funding costs affect their credit supply through a change in their expected net worth, and this sensitivity becomes stronger when banks are closer to the default threshold. This can also explain the nonlinearity of banks' credit-supply response to higher funding costs (cf. column 8 in Table 5), as the cost of external finance is convex with respect to the proximity to the default threshold (following the logic in Jensen and Meckling, 1976).

In column 3, we show a similar effect for low-liquidity banks, i.e., banks with a relatively low ratio of cash and central-bank reserves to total assets (measured again at the beginning of the sample period). This is consistent with the idea that banks' credit-supply response is

amplified when they cannot absorb the funding-cost increase and are, thus, more likely to experience a reduction in their expected net worth. In column 4, where we use a discrete variable based on the distribution of liquidity ratios, we see that the effect is driven primarily by high-liquidity banks lending disproportionately more.

In columns 5 and 6, we consider banks' share of non-performing loans (NPLs) out of total loans in the previous quarter. For both the continuous and the discrete version of the variable, with the latter capturing banks in the top tercile of the distribution, we find that high-NPL banks' lending response is positively related to their funding costs. This suggests that banks gamble for resurrection in the face of higher funding costs.

Robustness checks. We present a battery of robustness checks in the Online Appendix. In Table A.3, we show that our results are robust to controlling for *Deposit ratio transferred to* CDC_{bt-1}, the fraction T_{bt} of regulated deposits (no longer on bank b's balance sheet) transferred to the CDC over total liabilities of bank b in quarter $t-1$. This control accounts for the intermediary commissions that bank b receives in exchange for deposits transferred to the CDC to finance social housing (see Section 2.1.2). These commissions tend to be time-invariant and are therefore unlikely to covary with Gap_t.

In Table A.4, we show that our estimates are robust to different definitions of *Deposit* $ratio_{bt}$. Using the Banque de France's Cefit database, we can construct deposit ratios at the more granular bank-county level. The data are broken down by the same types of depositors as in the regulatory data, but cannot perfectly isolate regulated deposits. As such, we can only observe "special deposits," defined as regulated deposits plus ordinary savings.[20] In the first two columns, we re-run the same specifications as in columns 1 and 2 of Table 5, using as our exposure variable the special-deposit ratio at the bank-county-quarter level $bct-1$. The results are qualitatively similar, but the estimates are somewhat weaker. Any differences between the estimates in the first two columns and those in Table 5 do not stem

[20] In addition, bank liabilities are not fully observable in this more granular dataset. Thus, we use total deposits plus commercial paper as a proxy for total liabilities. We adjust deposit amounts for the percentage of deposits transferred to the CDC by using the same percentages as for the regulatory data. Let S_{bt} be the share of eligible deposits of bank b in quarter t, then: $Deposit\ ratio_{bct} = (S_{bt} \times (1 - T_{bt}) \times Special\ deposits_{bct} + (1 - S_{bt}) \times Special\ deposits_{bct})/Total\ liabilities_{bct}$. The data are available from Q1 2010 to Q4 2015.

from the definition of the deposit ratio employed in the latter table, however. To verify this, we re-run the same two regressions, and modify the bank-level deposit ratio according to the definition in the first two columns. The estimated coefficients on the relevant interaction term in Table A.4 are similar to those in Table 5.

Finally, we revisit the timing of our treatment-exposure variable, *Deposit ratio*$_{bt-1}$. We use lagged regulated-deposit ratios to ensure that our identifying variation does not stem from changes in the amount of regulated deposits but, rather, in the difference between the livret-A rate and the monetary-policy rate. We validate this by lagging *Deposit ratio*$_{bt-1}$ by another quarter, to *Deposit ratio*$_{bt-2}$, and re-running all regressions from Table 5. The results in Table A.5 are virtually unaltered, implying that changes in the quantity of regulated deposits cannot explain our findings. We provide additional evidence that a change in the quantity of regulated deposits in reaction to a change in their price is unlikely to affect our results by showing that (post-transfer) regulated deposits are barely sensitive to variation in the difference between the livret-A and the deposit facility rate. As can be seen in Figure B.3, the growth rate of banks' regulated deposits comoves weakly with the contemporaneous Gap_t.[21]

3.3 Reallocation of Credit

Bank loan-portfolio analysis. The change in credit supply we identify so far could mask an even larger credit reallocation if banks rebalance their portfolios toward higher-yielding loans so as to shield their profits (and ultimately net worth) from an increase in their funding costs. The credit registry does not record loan interest rates, so we cannot measure loan pricing directly. We instead measure reach for yield through the composition of banks' loan portfolios across categories that carry higher rates on average: smaller firms, firms in industries with more bankruptcies, riskier rated firms, and longer maturities. To test whether banks rebalance their portfolios in this way, we complement the credit registry with a bank-county-level dataset (Cefit) that provides more detailed information on the recipients

[21] This is even more so the case if one takes into account that the government increased the maximum amount of regulated deposits per person by 25% in Q4 2012 and another 20% in Q1 2013.

of credit, and additionally has credit information for non-corporate debtors, especially self-employed individuals (who are not covered in the credit registry).

Table 8: Reallocation of Credit: Bank-county-level Data

	Small firms Corporate loans	Small firms Total loans	Self-employed Total loans	High-bankruptcy industries Total loans	Risky firms Rated firms	MLT loans Total loans
	(1)	(2)	(3)	(4)	(5)	(6)
Deposit ratio × Gap	0.101*	0.040***	0.025***	0.397***	0.105*	0.118***
	(0.052)	(0.014)	(0.005)	(0.061)	(0.057)	(0.037)
Deposit ratio	-0.206**	-0.061*	0.005	-0.530***	-0.393***	-0.046
	(0.099)	(0.035)	(0.036)	(0.099)	(0.140)	(0.102)
Fixed Effects						
Bank-county	✓	✓	✓	✓	✓	✓
County × quarter	✓	✓	✓	✓	✓	✓
Banking group × quarter	✓	✓	✓	✓	✓	✓
Observations	25,550	25,500	25,500	25,086	24,080	25,500

Notes: The level of observation is all credit granted by bank b's branch(es) in county c in quarter t. The sample period is Q4 2010 to Q4 2015. The dependent variable in column 1 is the ratio of loans to small firms (with sales up to €1m) over corporate loans of bank b's branch(es) in county c in quarter t. The dependent variable in column 2 is the ratio of loans to small firms over total loans of bank b's branch(es) in county c in quarter t, winsorized at the 1st and 99th percentiles. The dependent variable in column 3 is the ratio of loans to self-employed individuals over total loans of bank b's branch(es) in county c in quarter t. The dependent variable in column 4 is the ratio of loans to firms in ~~(three-digit) industries with above-median occurrences of bankruptcies~~four-digit industries in the top tercile of bankruptcy rates over total loans of bank b's branch(es) in county c in quarter t. The dependent variable in column 5 is the ratio of loans to firms with a credit rating above 4 on the Banque de France's credit-rating scale (higher rating = closer to default) over all loans to rated firms (with balance-sheet data) granted by bank b's branch(es) in county c in quarter t. The dependent variable in column 6 is the ratio of medium- to long-term loans over total loans of bank b's branch(es) in county c in quarter t. $Deposit\ ratio_{bt-1}$ is the ratio of regulated deposits over total liabilities of bank b in quarter $t-1$. Gap_t is the difference between the rate on regulated deposits (livret A) and the ECB's deposit facility rate (in %) at the end of quarter t. Robust standard errors (clustered at the bank-county level) are in parentheses. $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$. The R-squared ranges from 0.66 to 0.96 across columns.

In Table 8, the level of observation is a bank-county-quarter *bct*, summarizing information on all branches of a given bank b in county c and quarter-year t. In columns 1 to 3, we estimate the adjustment of banks' loan portfolios across borrower types, using as dependent variables the shares of loans that go to each type. Following an increase in funding costs, affected banks raise the share of small firms (with sales ≤€1m) in their corporate loans (column 1; coefficient of 0.101, with a p-value of 0.052). This effect survives when we scale loans to small firms by banks' total loan portfolios, which comprise loans to all kinds of borrowers and not only the corporate lending captured by the credit registry (column 2). Affected banks also lend relatively more to self-employed individuals (column 3).[22]

To the extent that small firms make for potentially riskier borrowers than large firms, the increase in exposure to smaller borrowers suggests that banks facing higher funding

[22] This does not necessarily imply an increase in credit supply to small firms and self-employed individuals; instead, their relative importance in affected banks' loan portfolio increases.

costs take on more risk in search of higher yields. We provide further evidence of banks' risk taking in two ways. First, in column 4, we show that banks incurring higher average funding costs increase their exposure to firms with a higher risk of bankruptcy. For this purpose, we compute the ex-post bankruptcy probability at the industry level,[23] and use as our dependent variable the ratio of loans to firms in industries ~~with above-median occurrences of bankruptcies~~in the top tercile of bankruptcy rates over total loans.

Second, we exploit the credit ratings assigned by the Banque de France. To compute the proportion of loans accruing to risky firms, we label a firm as "risky" if it receives a rating worse than 4, which used to be the minimum rating required for a firm's loans to be eligible as collateral for the ECB (Cahn, Duquerroy, and Mullins, 2019). One drawback of this measure is that the Banque de France provides credit ratings only for firms with balance-sheet information.[24] Column 5 reports the result, and shows that regulated-deposit dependent banks increase their loan exposure to risky firms when their funding costs increase.

Our final test to study if banks reach for yield is to explore whether higher funding costs also induce banks to extend loans with a longer maturity. To this end, we compute the fraction of medium- to long-term loans in banks' loan portfolios and use it as dependent variable in column 6. The positive and significant coefficient on the interaction term $Deposit\ ratio_{bt-1} \times Gap_t$ confirms that banks increase the average maturity of their loan portfolios in reaction to higher average funding costs.

To quantify the extent of credit reallocation, we focus on corporate borrowers, the universe of which can be divided into large (safe) vs. small (risky) firms, representing, respectively, 43.8% and 56.3% of banks' average corporate-loan portfolios (see summary statistics in the first two rows of Panel C of Table 2). Column 1 of Table 8 implies that in response to a one-percentage-point increase in average funding costs, the average portfolio allocation changes to 33.6% and 66.4%, respectively. Using the estimated drop in net credit supply to firms of 8.4% up to 16.0% in response to the same funding-cost shock (from columns 1

[23] Based on additional data from the Banque de France (CCR) on bankruptcies and payment delinquencies, we use for each ~~(three-digit)~~four-digit industry the total number of such events and scale it by the number of firms (available in these data) in the respective industry.

[24] As such, we need to limit the denominator of the dependent variable to firms with sales of €750,000 or more.

and 2 of Table 5), we can decompose the reduction in credit supply to large firms into a proportional contraction of lending and a reallocation toward small firms.

If banks cut all corporate loans in proportion, credit to large firms would fall by 8.4% up to 16.0%. Their portfolio share falls to 33.6% instead, so credit to large firms drops by $(1 - 33.6\% \times (1 - 8.4\%)/43.8\% =)$ 29.7% up to $(1 - 33.6\% \times (1 - 16.0\%)/43.8\% =)$ 35.6%. The difference is the reallocation of banks' portfolios toward small firms. It accounts for $((43.8\% - 33.6\%) \times (1 - 16.0\%)/(43.8\% - 33.6\% \times (1 - 16.0\%)) =)$ 55% of the drop in credit to large firms with the estimate from column 2 of Table 5, and $((43.8\% - 33.6\%) \times (1 - 8.4\%)/(43.8\% - 33.6\% \times (1 - 8.4\%)) =)$ 72% with that from column 1. The remainder reflects the general contraction in lending.

Looking beyond banks' corporate-loan portfolio, our remaining estimates in Table 8 suggest even more credit reallocation across different borrowers, e.g., self-employed individuals. These significant reallocative effects, which can only be measured using granular credit-registry data, imply that analyses that focus on net credit supply alone likely underestimate the true effect of bank-level shocks on the real economy. In the extreme case, banks might be able to rebalance their loan portfolio enough to fully insulate their profits and, thus, ultimately their net worth. As a result, their net credit supply might not change at all, but the rebalancing can still have substantial real effects because in the presence of heterogeneous firms, credit reallocation affects aggregate productivity even when holding constant the overall amount of credit in the economy (e.g., Bau and Matray, 2023; Baqaee, Farhi, and Sangani, 2024).

Firm-level credit reallocation and geographic spillovers. Firms whose lenders cut credit may borrow from other banks, and the standard view is that large and less opaque firms are best placed to do so. To test this, we sum each firm's credit across all of its lenders and regress its log total credit on the credit-share-weighted average deposit ratio of these lenders interacted with Gap_t, with firm and county-by-industry-by-quarter fixed effects. Table A.8 shows a coefficient of -0.074 in the full sample (column 1), and a larger one for the firms usually thought most capable of switching lenders: large firms (-0.100

against −0.079 for small firms), firms with Banque de France balance-sheet data (−0.091 against −0.077), rated firms (−0.109 against −0.066), and firms with a long relationship with their main bank (−0.108 against −0.059). This pattern follows from the reallocation in Table 8. Exposed banks lend less and also move their corporate portfolio away from large firms toward small ones, so the borrowers they move away from bear a larger share of the cut even though they have better outside options.

We then ask whether firms in exposed areas make up for the cut by borrowing from lenders elsewhere. Following Beaumont, Matray, Tu, and Xu (2026), we add to the firm-level specification the credit-share-weighted mean deposit ratio of all firms in the firm's geography, in levels and interacted with Gap_t, where the geography is the city, the local labor market (employment zone), or the county (Table A.9), to allow for different levels of geographic spillovers. Compensation across space would make the coefficient on the geographic deposit ratio interacted with Gap_t positive. It is negative and significant at the 1% level in all three columns (−0.452, −0.421, and −0.357), and of similar size whether spillovers operate within cities, local labor markets, or counties, while the coefficient on the firm's own deposit ratio interacted with Gap_t stays between −0.210 and −0.173.

City-level analysis. Because the spillovers are of similar size whether we define geography by city, local labor market, or county, we assess whether banks' yield-seeking behavior is also reflected in the allocation of credit at a more aggregate level by aggregating all our variables at the city (ZIP code) level. To do so, we compute a weighted average of bank dependence on regulated deposits to provide us with a city-level credit shock, and treat all cities as small independent economies facing an "aggregate shock." This type of geographical approach is designed to capture "semi-aggregate" effects (e.g., Greenstone, Max, and Nguyen, 2020).[25]

To construct the city-wide shock, we use a shift-share approach by considering the funding structure of all banks lending to firms in a given ZIP code. Namely, for each bank b lending to firms f in ZIP code k,[26] we weight the bank-level deposit ratio by the respective bank b's

[25] These are "semi-aggregate" because while they measure the change in credit at the city level, by construction these are still reduced-form elasticities estimated relative to the control group that is assumed to be unaffected.

[26] There are around 33,000 distinct cities in France, each belonging to only one county.

lagged share of all loans in ZIP code k:

$$Deposit\ ratio_{kt} = \sum_{f \in k} \frac{Credit_{fbt-1}}{\sum_{f \in k} Credit_{fbt-1}} Deposit\ ratio_{bt}, \tag{2}$$

where $Credit_{fbt-1}$ measures the euro amount of debt outstanding between firm f and (all branches of) bank b in quarter $t-1$, and $Deposit\ ratio_{bt}$ is the ratio of regulated deposits over total liabilities of bank b in quarter t.

We then estimate the following specification at the ZIP-code-quarter level kt:

$$y_{kt} \quad = \quad \beta_1 Deposit\ ratio_{kt-1} \times Gap_t + \beta_2 Deposit\ ratio_{kt-1} + \psi_{ct} + \delta_k + \epsilon_{kt}, \tag{3}$$

where y_{kt} is a variable based on the cross-section of loans granted to firms in ZIP code k in quarter t, and ψ_{ct} and δ_k denote county-quarter and ZIP-code fixed effects, respectively. Standard errors are clustered at the ZIP-code level.

As such, credit outcomes reflect not only the intensive margin of lending, as in our bank-firm-level specification, but also its extensive margin. β_1 therefore captures reduced lending to first-time borrowers in cities whose firms borrow from regulated-deposit dependent banks, but not the reshuffling of existing borrowers across banks within a city. While a higher level of aggregation allows us to estimate whether firms are able to substitute credit across differentially affected local lenders, it prevents us, by construction, from controlling for time-varying unobserved heterogeneity at the firm level. To keep cities comparable, we include county-by-time fixed effects, so that we compare only cities within the same county. Such a strategy removes time-varying unobserved heterogeneity across counties, such as differences in credit demand, in business cycles and dynamism, or in industrial composition that may influence our estimates.

Table 9 reports the results for both the net supply of credit and its composition at the city level.[27] In column 1, we estimate eq. (3) and use the natural logarithm of total (corporate) credit as dependent variable. We find a large negative coefficient, significant at the 1%

[27] As for this exercise we require data on loan recipients' cities, all dependent variables are based on corporate-lending data from the credit registry.

Table 9: Aggregate Credit Effects of Shocks to Banks' Funding Costs

Sample	ln(Total credit) All (1)	Large firms / Total credit All (2)	High-bankruptcy ind. / Total credit All (3)	MLT credit / Total credit All (4)	ln(Total credit) > 5 firms (5)	Large firms / Total credit > 5 firms (6)	High-bankruptcy ind. / Total credit > 5 firms (7)	MLT credit / Total credit > 5 firms (8)
Deposit ratio × Gap	-0.175***	-0.036***	0.103***	0.060***	-0.338***	-0.060**	0.216***	0.044***
	(0.053)	(0.014)	(0.019)	(0.016)	(0.056)	(0.025)	(0.025)	(0.015)
Deposit ratio	-0.008	-0.121***	-0.302***	0.189***	-1.053***	-0.382***	-0.361***	0.008
	(0.107)	(0.025)	(0.039)	(0.031)	(0.137)	(0.054)	(0.053)	(0.030)
Fixed Effects								
ZIP-code	✓	✓	✓	✓	✓	✓	✓	✓
County-quarter	✓	✓	✓	✓	✓	✓	✓	✓
Observations	673,947	673,947	672,472	673,947	344,626	344,626	344,626	344,626

Notes: The level of observation is the ZIP-code-quarter level kt. The sample period is Q4 2010 to Q4 2015. In the last four columns, the sample is limited to ZIP codes with more than five firms (with records in the credit registry). The dependent variable in columns 1 and 5 is the natural logarithm of the total euro amount of debt outstanding of all firms in ZIP code k in quarter t. The dependent variable in columns 2 and 6 is the ratio of all loans accruing to large firms (with sales in excess of €1m; firms without FIBEN balance-sheet data are classified as small) over the total euro amount of debt outstanding of all firms in ZIP code k in quarter t. The dependent variable in columns 3 and 7 is the ratio of all loans accruing to firms in ~~(three-digit) industries with above-median occurrences of bankruptcies~~four-digit industries in the top tercile of bankruptcy rates over the total euro amount of debt outstanding of all firms in ZIP code k in quarter t. The dependent variable in columns 4 and 8 is the ratio of all medium- to long-term loans over the total euro amount of debt outstanding of all firms in ZIP code k in quarter t. *Deposit ratio*$_{kt-1}$ is the loan-exposure-weighted average *Deposit ratio*$_{bt-1}$ of the lenders to all firms in ZIP code k in quarter $t-1$ (see eq. (2)), where *Deposit ratio*$_{bt-1}$ is the ratio of regulated deposits over total liabilities of bank b in quarter $t-1$. Gap_t is the difference between the rate on regulated deposits (livret A) and the ECB's deposit facility rate (in %) at the end of quarter t. Robust standard errors (clustered at the ZIP-code level) are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The R-squared ranges from 0.69 to 0.97 across columns.

level, implying that non-affected banks cannot perfectly compensate for the change in credit supply from affected banks.

Deposit ratio$_{kt-1}$ × *Gap*$_t$ captures the difference in the city-level weighted average funding costs of regulated-deposit dependent banks in relationships with firms in the respective cities vs. cities that are home to firms only in relationships with otherwise-funded banks. Our estimate in column 1 implies that cities see their credit drop by 17.5% if they face funding costs that are one percentage point higher, which is economically significant and slightly larger than the corresponding effect at the bank-firm level (−0.160 in column 2 of Table 5). Besides indicating a limited role for the extensive margin of lending, this suggests that borrowers have imperfect ability to switch banks so as to smooth over credit-supply shocks, consistent with the existence of sticky lending relationships.[28]

In column 2 of Table 9, we use as dependent variable the fraction of loans to large vs. all firms. Consistent with affected banks' loan-portfolio rebalancing (in Table 8), we again find a large negative effect, significant at the 1% level. At the city level, credit contraction following adverse shocks to banks' funding costs affects primarily large rather than small firms.

In columns 3 and 4, we consider two additional dimensions of cross-sectional heterogeneity implied by Table 8, the respective dependent variables of which we can compute at the aggregate city level based on the credit-registry data. In column 3, we show that greater risk taking in terms of lending to firms in risky industries (cf. column 4 of Table 8) also holds at the more aggregate level. Similarly, in column 4, we find that affected banks' extension of longer-term loans (cf. column 6 of Table 8) is also reflected in our more aggregate estimates. All of these estimates are similar or larger after removing ZIP codes with at most five firms (with records in the credit registry) in the last four columns of Table 9.

[28] Several factors can raise switching costs: the existence of a "stigma" when switching (e.g., Darmouni, 2020) or the lack of geographic diversification across banks (e.g., Célérier and Matray, 2019). For a discussion of the importance of comparing firm-level and more aggregate estimates, see, for instance, Chodorow-Reich (2014) and Greenstone, Max, and Nguyen (2020).

3.4 Firm-level Real Effects

Our city-level results show that a reduction in the supply of credit by regulated-deposit dependent banks during periods in which they have to pay higher interest on these deposits is not compensated for by an increase in the supply of credit by otherwise-funded banks. At the firm level, this imperfect ability to substitute credit across banks is further exacerbated by the fact that the small and medium-sized firms in our sample cannot compensate for a change in bank credit (at least in the short run) with other types of financing, as 99% of them do not have any capital-market financing. As a result, variation in banks' funding costs should have real effects.

First, we show that our effects on bank lending in Table 5 are present in the subsample of firms with balance-sheet data available, which roughly corresponds to the group of large firms (FIBEN covers firms with sales of €750,000 or more) in the credit registry. In Table A.6, we re-run the same specifications as in Table 5 on this sample, and find that all credit-based results continue to hold and are even stronger than in the overall sample.

To test for the real effects of banks' credit-supply response to variation in their average funding costs, we estimate regressions at the firm-year level. We use a shift-share approach similar to eq. (2). To compute firm-level exposure to credit-supply shocks, measured by the variable $Deposit\ ratio_{ft}$, we use for each lender to firm f their bank-level deposit ratio, and weight the latter by the lagged share of all loans granted to firm f by bank b's branch(es) in county c.

We then estimate the following regression specification at the firm-year level ft:

$$y_{ft} = \beta_1 Deposit\ ratio_{ft-1} \times Gap_t + \beta_2 Deposit\ ratio_{ft-1} + \psi_{ci(f)t} + \delta_f + \epsilon_{ft}, \quad (4)$$

where y_{ft} is an outcome of firm f in year t, and $\psi_{ci(f)t}$ and δ_f denote firm f's county-industry-year and firm fixed effects, respectively. Standard errors are clustered at the firm level.

In Table 10, we estimate eq. (4) and use multiple firm-level outcomes. We find that more exposed firms see a drop in their total capital (column 1), mostly driven by a drop in physical

capital (property, plant, and equipment, column 2). In terms of economic magnitude, *Deposit ratio*$_{ft-1}$ × *Gap*$_t$ captures the difference in the weighted average funding costs of regulated-deposit dependent banks that a firm is in a relationship with, as opposed to firms that are only in relationships with otherwise-funded banks. Therefore, our estimate in column 1 implies that firms see a drop in their stock of total capital by ~~3.5~~3.8% if their relationship banks incur funding costs that are one percentage point higher.[29]

Table 10: Firm-level Real Effects of Shocks to Banks' Funding Costs

	ln(Total capital) (1)	ln(PP&E) (2)	$\frac{\text{CapEx}}{\text{Total capital}}$ (3)	$\frac{\text{Tangible investment}}{\text{PP\&E}}$ (4)	ln(Employment) (5)
Deposit ratio × Gap	-0.038**	-0.050***	-0.057**	-0.029**	-0.015
	(0.015)	(0.016)	(0.024)	(0.015)	(0.011)
Deposit ratio	0.167***	0.211***	0.034	0.018	0.032
	(0.030)	(0.032)	(0.042)	(0.026)	(0.022)
Fixed Effects					
Firm	✓	✓	✓	✓	✓
County-ind.-yr.	✓	✓	✓	✓	✓
Observations	423,630	423,630	423,630	423,630	423,630

Notes: The level of observation is the firm-year level ft. Furthermore, the sample is limited to rated firms (with available balance-sheet data). The sample period is 2010 to 2015. All dependent variables are measured at the firm-year level ft. *CapEx*$_{ft}$ is computed as the sum of firm f's tangible and intangible investment in year t. *Deposit ratio*$_{ft-1}$ is the loan-exposure-weighted average *Deposit ratio*$_{bt-1}$ of all bank branches lending to firm f in quarter $t-1$, where *Deposit ratio*$_{bt-1}$ is the ratio of regulated deposits over total liabilities of bank b in quarter $t-1$. *Gap*$_t$ is the difference between the rate on regulated deposits (livret A) and the ECB's deposit facility rate (in %) at the end of quarter t. Industry fixed effects are defined at the three-digit level. Robust standard errors (clustered at the firm level) are in parentheses. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Missing values of the outcome variables are set to zero. The R-squared ranges from ~~0.42~~0.41 to 0.97 across columns.

In columns 3 and 4, we show that the relative reduction in total and physical capital in columns 1 and 2 is because more exposed firms actively reduce their investment efforts, as measured by capital expenditure over total capital (column 3) and tangible investment over physical capital (column 4). Finally, we also estimate a negative, albeit insignificant, effect on employment in column 5.

[29] Our sample consists of the 423,630 firm-years with nonmissing investment over lagged total assets. Missing values of the other outcomes are set to zero, so that every column uses the same firm-years. Leaving them missing instead reduces the sample to between 416,532 and 420,366 firm-years depending on the column, and yields similar estimates in Table A.10 of the Online Appendix.

4 Conclusion

Banks incur fixed operating expenses to maintain a deposit franchise, but there is a lively debate about the extent to which banks' deposit franchise matters for bank behavior (e.g., Drechsler, Savov, and Schnabl, 2021; Begenau and Stafford, 2022). We show that a higher operating cost of maintaining a deposit franchise reduces banks' credit supply to the real sector.

To obtain shifts in these costs that banks do not control, we exploit the existence of regulated deposits offered by all banks in France whose rates are set by the government and not by the banks themselves. As banks' total fixed costs of operation translate into fluctuations in bank profits, we can estimate an elasticity of banks' credit supply with respect to their equity. The elasticity is 0.67, and we also uncover a highly nonlinear response as banks barely adjust their lending to small increases in their funding costs and cut it more for larger increases. Consistent with the idea that changes in banks' funding or operating costs affect their credit supply through changes in their probability of default that depress their expected net worth, the credit-supply response is stronger for weakly capitalized banks and for banks with lower liquidity buffers.

Banks adjust the composition of their credit as well as its net supply. To insulate their profits, they reach for yield by rebalancing their loan portfolio toward smaller and riskier firms, and longer-term loans. This implies that even when the quantity of loans supplied in the aggregate remains stable, banks' operating costs can affect aggregate output if borrowers and projects exhibit different productivity. How the joint distribution of banks' and borrowers' heterogeneity shapes the transmission of bank-level shocks to aggregate output remains an open question.

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ONLINE APPENDIX

A Supplementary Tables

Table A.1: Average Effect of Funding Costs on Credit Supply: Time-invariant Deposit Ratio I

	ln(Credit) (1)	ln(Credit) (2)	ln(Credit) (3)	ln(Credit) (4)	ln(Credit) (5)	ln(Credit) (6)	ln(Credit) (7)	ln(Credit) (8)
Deposit ratio × Gap	-0.078***	-0.187***	-0.210***	-0.190***	-0.181***	-0.186***	-0.180***	
	(0.028)	(0.047)	(0.051)	(0.047)	(0.046)	(0.046)	(0.047)	
Total deposit ratio × Gap			0.023					
			(0.020)					
$\frac{\text{Loans to households}}{\text{Total loans}}$ × Gap				0.013				
				(0.018)				
$\frac{\text{Loans to households}}{\text{Total loans}}$				-0.021				
				(0.043)				
Equity ratio × Gap					0.120		0.122	
					(0.181)		(0.184)	
Equity ratio					0.292		0.279	
					(0.567)		(0.522)	
Bank size × Gap						0.000	0.000	
						(0.002)	(0.002)	
Bank size						-0.008	-0.002	
						(0.040)	(0.041)	
Deposit ratio × Gap in top tercile								-0.156**
								(0.060)
Deposit ratio × Gap in 2nd tercile								-0.038
								(0.044)
Fixed Effects								
Firm-bank-county	✓	✓	✓	✓	✓	✓	✓	✓
Firm-quarter	✓	✓	✓	✓	✓	✓	✓	✓
County-quarter	✓	✓	✓	✓	✓	✓	✓	✓
BHC-quarter		✓	✓	✓	✓	✓	✓	✓
Observations	4,495,354	4,495,354	4,495,354	4,495,354	4,495,354	4,495,354	4,495,354	4,495,354

Notes: The level of observation is credit to firm f by bank b's branch(es) in county c in quarter t. The sample period is Q4 2010 to Q4 2015. The dependent variable is the natural logarithm of the euro amount of debt outstanding between firm f and bank b's branch(es) in county c in quarter t. *Deposit ratio*$_b$ is the (time-invariant) ratio of regulated deposits over total liabilities of bank b in Q3 2010. *Total deposit ratio*$_{bt}$ is the ratio of all deposits over total liabilities of bank b in Q3 2010. $\frac{\textit{Loans to households}}{\textit{Total loans}}_{bt-1}$ is the fraction of household lending out of total lending by bank b in quarter $t-1$. *Equity ratio*$_{bt-1}$ is the ratio of equity over total assets of bank b in quarter $t-1$. *Bank size*$_{bt-1}$ is the natural logarithm of total assets of bank b in quarter $t-1$. *Gap*$_t$ is the difference between the rate on regulated deposits (livret A) and the ECB's deposit facility rate (in %) at the end of quarter t. *Gap in top* (2nd) *tercile*$_t$ is a dummy variable for whether *Gap*$_t$ ranges in the top (middle) tercile of its distribution. Robust standard errors (clustered at the bank level) are in parentheses. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The R-squared is 0.94 in all columns.

Table A.2: Average Effect of Funding Costs on Credit Supply: Time-invariant Deposit Ratio II

	ln(Credit) (1)	ln(Credit) (2)	ln(Credit) (3)	ln(Credit) (4)	ln(Credit) (5)	ln(Credit) (6)	ln(Credit) (7)	ln(Credit) (8)
Deposit ratio × Gap	-0.079***	-0.174***	-0.191***	-0.174***	-0.169***	-0.172***	-0.167***	
	(0.028)	(0.046)	(0.051)	(0.047)	(0.045)	(0.046)	(0.045)	
Total deposit ratio × Gap			0.019					
			(0.021)					
$\frac{\text{Loans to households}}{\text{Total loans}}$ × Gap				0.008				
				(0.019)				
$\frac{\text{Loans to households}}{\text{Total loans}}$				-0.012				
				(0.045)				
Equity ratio × Gap					0.198		0.205	
					(0.188)		(0.187)	
Equity ratio					0.160		0.134	
					(0.574)		(0.532)	
Bank size × Gap						0.001	0.001	
						(0.002)	(0.002)	
Bank size						-0.009	-0.003	
						(0.040)	(0.040)	
Deposit ratio × Gap in top tercile								-0.143**
								(0.059)
Deposit ratio × Gap in 2nd tercile								-0.032
								(0.043)
Fixed Effects								
Firm-bank-county	✓	✓	✓	✓	✓	✓	✓	✓
Firm-quarter	✓	✓	✓	✓	✓	✓	✓	✓
County-quarter	✓	✓	✓	✓	✓	✓	✓	✓
BHC-quarter		✓	✓	✓	✓	✓	✓	✓
Observations	4,495,354	4,495,354	4,495,354	4,495,354	4,495,354	4,495,354	4,495,354	4,495,354

Notes: The level of observation is credit to firm f by bank b's branch(es) in county c in quarter t. The sample period is Q4 2010 to Q4 2015. The dependent variable is the natural logarithm of the euro amount of debt outstanding between firm f and bank b's branch(es) in county c in quarter t. *Deposit ratio*$_b$ is the (time-invariant) ratio of regulated deposits over total liabilities of bank b in Q4 2010. *Total deposit ratio*$_{bt}$ is the ratio of all deposits over total liabilities of bank b in Q4 2010. $\frac{\textit{Loans to households}}{\textit{Total loans}}_{bt-1}$ is the fraction of household lending out of total lending by bank b in quarter $t-1$. *Equity ratio*$_{bt-1}$ is the ratio of equity over total assets of bank b in quarter $t-1$. *Bank size*$_{bt-1}$ is the natural logarithm of total assets of bank b in quarter $t-1$. *Gap*$_t$ is the difference between the rate on regulated deposits (livret A) and the ECB's deposit facility rate (in %) at the end of quarter t. *Gap in top* (2nd) *tercile*$_t$ is a dummy variable for whether *Gap*$_t$ ranges in the top (middle) tercile of its distribution. Robust standard errors (clustered at the bank level) are in parentheses. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The R-squared is 0.94 in all columns.

Table A.3: Average Effect of Funding Costs on Lending by Deposit-funded Banks: Control for Income from CDC Transfer

	ln(Credit) (1)	ln(Credit) (2)	ln(Credit) (3)	ln(Credit) (4)	ln(Credit) (5)	ln(Credit) (6)	ln(Credit) (7)	ln(Credit) (8)
Deposit ratio × Gap	-0.063**	-0.130***	-0.121**	-0.129***	-0.131***	-0.131***	-0.131***	
	(0.026)	(0.043)	(0.050)	(0.044)	(0.041)	(0.045)	(0.042)	
Deposit ratio	0.309**	0.294*	0.390**	0.297*	0.261	0.274**	0.257*	0.161
	(0.120)	(0.164)	(0.186)	(0.162)	(0.158)	(0.135)	(0.139)	(0.167)
Deposit ratio transferred to CDC	-0.406***	-0.301**	-0.226*	-0.303**	-0.302**	-0.310**	-0.310**	-0.321**
	(0.113)	(0.125)	(0.128)	(0.125)	(0.126)	(0.127)	(0.127)	(0.125)
Total deposit ratio			-0.181*					
			(0.098)					
Total deposit ratio × Gap			0.018					
			(0.023)					
$\frac{\text{Loans to households}}{\text{Total loans}}$ × Gap				0.008				
				(0.019)				
$\frac{\text{Loans to households}}{\text{Total loans}}$				-0.010				
				(0.045)				
Equity ratio × Gap					0.235		0.250	
					(0.194)		(0.187)	
Equity ratio					0.266		0.165	
					(0.571)		(0.537)	
Bank size × Gap						0.002	0.002	
						(0.002)	(0.002)	
Bank size						-0.019	-0.013	
						(0.041)	(0.041)	
Deposit ratio × Gap in top tercile								-0.090**
								(0.046)
Deposit ratio × Gap in 2nd tercile								0.002
								(0.031)
Fixed Effects								
Firm-bank-county	✓	✓	✓	✓	✓	✓	✓	✓
Firm-quarter	✓	✓	✓	✓	✓	✓	✓	✓
County-quarter	✓	✓	✓	✓	✓	✓	✓	✓
BHC-quarter		✓	✓	✓	✓	✓	✓	✓
Observations	4,495,354	4,495,354	4,495,354	4,495,354	4,495,354	4,495,354	4,495,354	4,495,354

Notes: The R-squared is 0.94 in all columns.

Table A.4: Average Effect of Funding Costs on Lending by Deposit-funded Banks: Robustness

Deposits	ln(Credit) Regulated deposits + ordinary savings (branch level)		ln(Credit) Regulated deposits + ordinary savings (bank level)	
	(1)	(2)	(3)	(4)
Deposit ratio × Gap	-0.027	-0.053**	-0.061***	-0.117***
	(0.022)	(0.024)	(0.024)	(0.037)
Deposit ratio	0.033	0.065	0.076	0.104
	(0.048)	(0.051)	(0.081)	(0.096)
Fixed Effects				
Firm-bank-county	✓	✓	✓	✓
Firm-quarter	✓	✓	✓	✓
County-quarter	✓	✓	✓	✓
BHC-quarter		✓		✓
Observations	5,127,382	5,127,382	4,495,354	4,495,354

Notes: The level of observation is credit to firm f by bank b's branch(es) in county c in quarter t. The sample period is Q1 2010 to Q4 2015 in the first two columns, and Q4 2010 to Q4 2015 in the last two columns. The dependent variable is the natural logarithm of the euro amount of debt outstanding between firm f and bank b's branch(es) in county c in quarter t. In the first two columns, $Deposit\ ratio_{bct-1}$ is the ratio of regulated deposits plus ordinary savings accounts over total deposits and commercial paper of bank b's branch(es) in county c in quarter $t-1$. In the last two columns, $Deposit\ ratio_{bt-1}$ is the ratio of regulated deposits plus ordinary savings accounts over total liabilities of bank b in quarter $t-1$. Gap_t is the difference between the rate on regulated deposits (livret A) and the ECB's deposit facility rate (in %) at the end of quarter t. Robust standard errors (clustered at the bank level) are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The R-squared is 0.94 in all columns.

Table A.5: Average Effect of Funding Costs on Lending by Deposit-funded Banks: Robustness to Timing

	ln(Credit) (1)	ln(Credit) (2)	ln(Credit) (3)	ln(Credit) (4)	ln(Credit) (5)	ln(Credit) (6)	ln(Credit) (7)	ln(Credit) (8)
Deposit ratio$_{t-2}$ × Gap	-0.087***	-0.159***	-0.134**	-0.158***	-0.157***	-0.159***	-0.156***	
	(0.027)	(0.043)	(0.052)	(0.043)	(0.042)	(0.043)	(0.042)	
Deposit ratio$_{t-2}$	0.153*	0.195	0.347**	0.200	0.172	0.191*	0.181*	0.024
	(0.088)	(0.128)	(0.164)	(0.127)	(0.117)	(0.102)	(0.100)	(0.120)
Total deposit ratio$_{t-2}$ × Gap			0.027					
			(0.023)					
Total deposit ratio$_{t-2}$			-0.208**					
			(0.088)					
$\frac{\text{Loans to households}}{\text{Total loans}}$ × Gap				0.011				
				(0.019)				
$\frac{\text{Loans to households}}{\text{Total loans}}$				-0.020				
				(0.045)				
Equity ratio × Gap					0.253		0.265	
					(0.196)		(0.192)	
Equity ratio					0.513		0.527	
					(0.647)		(0.552)	
Bank size × Gap						0.001	0.001	
						(0.002)	(0.002)	
Bank size						-0.011	0.003	
						(0.038)	(0.036)	
Deposit ratio$_{t-2}$ × Gap in top tercile								-0.127**
								(0.049)
Deposit ratio$_{t-2}$ × Gap in 2nd tercile								-0.011
								(0.030)
Fixed Effects								
Firm-bank-county	✓	✓	✓	✓	✓	✓	✓	✓
Firm-quarter	✓	✓	✓	✓	✓	✓	✓	✓
County-quarter	✓	✓	✓	✓	✓	✓	✓	✓
BHC-quarter		✓	✓	✓	✓	✓	✓	✓
Observations	4,303,804	4,303,804	4,303,804	4,303,804	4,303,804	4,303,804	4,303,804	4,303,804

Notes: The level of observation is credit to firm f by bank b's branch(es) in county c in quarter t. The sample period is Q1 2011 to Q4 2015. The dependent variable is the natural logarithm of the euro amount of debt outstanding between firm f and bank b's branch(es) in county c in quarter t. *Deposit ratio*$_{bt-2}$ is the ratio of regulated deposits over total liabilities of bank b in quarter $t-2$. *Total deposit ratio*$_{bt-2}$ is the ratio of all deposits over total liabilities of bank b in quarter $t-2$. $\frac{\text{Loans to households}}{\text{Total loans}}_{bt-1}$ is the fraction of household lending out of total lending by bank b in quarter $t-1$. *Equity ratio*$_{bt-1}$ is the ratio of equity over total assets of bank b in quarter $t-1$. *Bank size*$_{bt-1}$ is the natural logarithm of total assets of bank b in quarter $t-1$. Gap_t is the difference between the rate on regulated deposits (livret A) and the ECB's deposit facility rate (in %) at the end of quarter t. *Gap in top* (2nd) *tercile*$_t$ is a dummy variable for whether Gap_t ranges in the top (middle) tercile of its distribution. Robust standard errors (clustered at the bank level) are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The R-squared is 0.94 in all columns.

Table A.6: Average Effect of Funding Costs on Lending by Deposit-funded Banks: Firms with Balance-sheet Data

	ln(Credit) (1)	ln(Credit) (2)	ln(Credit) (3)	ln(Credit) (4)	ln(Credit) (5)	ln(Credit) (6)	ln(Credit) (7)	ln(Credit) (8)
Deposit ratio × Gap	-0.086**	-0.192***	-0.168**	-0.192***	-0.188***	-0.194***	-0.188***	
	(0.036)	(0.054)	(0.065)	(0.057)	(0.055)	(0.055)	(0.055)	
Deposit ratio	0.150	0.278	0.412*	0.273	0.224	0.207	0.177	0.094
	(0.143)	(0.198)	(0.241)	(0.197)	(0.184)	(0.149)	(0.148)	(0.202)
Total deposit ratio × Gap			0.015					
			(0.030)					
Total deposit ratio			-0.207					
			(0.128)					
$\frac{\text{Loans to households}}{\text{Total loans}}$ × Gap				-0.002				
				(0.025)				
$\frac{\text{Loans to households}}{\text{Total loans}}$				0.016				
				(0.062)				
Equity ratio × Gap					0.262		0.316	
					(0.280)		(0.263)	
Equity ratio					0.320		0.019	
					(0.638)		(0.591)	
Bank size × Gap						0.003	0.004	
						(0.003)	(0.003)	
Bank size						-0.041	-0.035	
						(0.056)	(0.057)	
Deposit ratio × Gap in top tercile								-0.148**
								(0.062)
Deposit ratio × Gap in 2nd tercile								-0.025
								(0.046)
Fixed Effects								
Firm-bank-county	✓	✓	✓	✓	✓	✓	✓	✓
Firm-quarter	✓	✓	✓	✓	✓	✓	✓	✓
County-quarter	✓	✓	✓	✓	✓	✓	✓	✓
BHC-quarter		✓	✓	✓	✓	✓	✓	✓
Observations	1,926,745	1,926,745	1,926,745	1,926,745	1,926,745	1,926,745	1,926,745	1,926,745

Notes: The level of observation is credit to firm f by bank b's branch(es) in county c in quarter t. Furthermore, the sample is limited to firms with available balance-sheet data. The sample period is Q4 2010 to Q4 2015. The dependent variable is the natural logarithm of the euro amount of debt outstanding between firm f and bank b's branch(es) in county c in quarter t. *Deposit ratio*$_{bt-1}$ is the ratio of regulated deposits over total liabilities of bank b in quarter $t-1$. *Total deposit ratio*$_{bt-1}$ is the ratio of all deposits over total liabilities of bank b in quarter $t-1$. $\frac{\textit{Loans to households}}{\textit{Total loans}}_{bt-1}$ is the fraction of household lending out of total lending by bank b in quarter $t-1$. *Equity ratio*$_{bt-1}$ is the ratio of equity over total assets of bank b in quarter $t-1$. *Bank size*$_{bt-1}$ is the natural logarithm of total assets of bank b in quarter $t-1$. Gap_t is the difference between the rate on regulated deposits (livret A) and the ECB's deposit facility rate (in %) at the end of quarter t. *Gap in top* (2nd) *tercile*$_t$ is a dummy variable for whether Gap_t ranges in the top (middle) tercile of its distribution. Robust standard errors (clustered at the bank level) are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The R-squared is 0.92 in all columns.

Table A.7: Average Effect of Funding Costs on Credit Supply: Bank Controls Interacted with Quarter Fixed Effects

Dependent variable	ln(Credit)				
	(1)	(2)	(3)	(4)	(5)
Deposit ratio × Gap	-0.160***	-0.142***	-0.166***	-0.136***	-0.128***
	(0.043)	(0.048)	(0.045)	(0.044)	(0.043)
Deposit ratio	0.137	0.297*	0.152	0.136	0.368**
	(0.137)	(0.177)	(0.138)	(0.137)	(0.179)
Control × quarter					
Total deposits / liabilities		✓			✓
Liquidity ratio			✓		✓
Profits / revenues$_{t-1}$				✓	✓
Fixed Effects					
Firm × bank × county	✓	✓	✓	✓	✓
Firm × quarter	✓	✓	✓	✓	✓
County × quarter	✓	✓	✓	✓	✓
Banking group × quarter	✓	✓	✓	✓	✓
Observations	4,495,354	4,495,354	4,495,354	4,495,035	4,495,035

Notes: The level of observation is credit to firm f by bank b's branch(es) in county c in quarter t. The sample period is Q4 2010 to Q4 2015, and each column requires only its own controls: columns 4 and 5 drop 189 observations without prior-year profits over revenues and 130 fixed-effect singletons. *Deposit ratio*$_{bt-1}$ is the ratio of regulated deposits over total liabilities of bank b in quarter $t-1$, and *Gap*$_t$ is the difference between the rate on regulated deposits (livret A) and the ECB's deposit facility rate (in %) at the end of quarter t. Each checked control enters interacted with a full set of quarter indicators, which spans its level; the quarter-specific slopes are not reported. *Total deposits / liabilities*$_{bt-1}$ is not winsorized. Profits over revenues use the prior-year annual value, held constant within the year; liquidity and total deposits over liabilities retain their quarterly timing. Liquidity and profits over revenues are winsorized at the 1st and 99th percentiles. Robust standard errors (clustered at the bank level) are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.8: Total Firm Credit by Initial Firm Characteristics

Dependent variable	ln(Total firm credit)								
Split	Full sample	Size		FIBEN		Rating		Relationship	
Group	All	Large	Small	In	Outside	Rated	Unrated	Long	Short
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Deposit ratio × Gap	-0.074***	-0.100***	-0.079***	-0.091**	-0.077***	-0.109***	-0.066***	-0.108***	-0.059***
	(0.010)	(0.038)	(0.010)	(0.036)	(0.010)	(0.020)	(0.011)	(0.016)	(0.012)
Fixed Effects									
Firm	✓	✓	✓	✓	✓	✓	✓	✓	✓
County × industry × quarter	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	10,162,977	1,299,598	8,799,456	1,437,629	8,661,444	3,092,193	7,009,652	5,384,222	4,717,379

Notes: The level of observation is the firm-quarter level ft. The dependent variable is the natural logarithm of firm f's total committed credit, drawn plus undrawn, summed across all of its banks. *Deposit ratio*$_{ft-1}$ is the average of its lenders' *Deposit ratio*$_{bt-1}$, weighted by their shares of the firm's credit in quarter $t-1$; every regression also includes it in levels. Unmatched bank records contribute zero. Each column is a separate unweighted regression. Groups are fixed at the firm's first sample quarter, with annual sales and FIBEN coverage measured in that quarter's calendar year. Large firms have annual sales of at least €1m; firms without FIBEN data are classified as small. FIBEN indicates availability of a Banque de France balance sheet; rated indicates an observed nonzero credit rating. Relationship length is the number of quarters since the first observed link to the firm's initially largest lender: long is above the median of 9 quarters, short is at or below it; 300,829 of 754,110 initial main-bank relationships are left-censored. Singletons are removed separately within each group, so split-column counts need not sum to the full-sample count. Size and FIBEN coverage overlap. Robust standard errors (clustered at the firm level) are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.9: Total Firm Credit and Geographic Exposure

Dependent variable	ln(Total firm credit)		
Geography	Commune	Employment zone	Department
	(1)	(2)	(3)
Own deposit ratio × Gap	-0.210***	-0.173***	-0.199***
	(0.021)	(0.037)	(0.037)
Geographic deposit ratio × Gap	-0.452***	-0.421***	-0.357***
	(0.038)	(0.118)	(0.113)
Own deposit ratio	0.275***	0.045	0.026
	(0.047)	(0.084)	(0.074)
Geographic deposit ratio	0.001	0.227	0.335
	(0.077)	(0.284)	(0.254)
Fixed Effects			
Firm	✓	✓	✓
Quarter	✓	✓	✓
Observations	9,504,142	9,504,142	9,504,142

Notes: The level of observation is the firm-quarter level ft. The dependent variable is the natural logarithm of firm f's total committed credit, drawn plus undrawn, summed across all of its banks. *Own deposit ratio*$_{ft-1}$ is the average of its lenders' *Deposit ratio*$_{bt-1}$, weighted by their shares of the firm's credit in quarter $t-1$. *Geographic deposit ratio* is the mean of *Own deposit ratio*$_{ft-1}$ across firms in the firm's current-quarter geography (city, employment zone, or county), weighted by firms' lagged credit shares; it includes the firm's own contribution. The same shares are the regression weights. The sample contains firms with positive credit in consecutive quarters and at least five eligible firms in their city-quarter, and is identical across columns. Unmatched bank ratios contribute zero. Robust standard errors (clustered at the indicated geography) are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.10: Firm-level Real Effects of Shocks to Banks' Funding Costs: Missing Outcomes Left Missing

	ln(Total capital)	ln(PP&E)	$\frac{\text{CapEx}}{\text{Total capital}}$	$\frac{\text{Tangible investment}}{\text{PP\&E}}$	ln(Employment)
	(1)	(2)	(3)	(4)	(5)
Deposit ratio × Gap	-0.040***	-0.043***	-0.054**	-0.028*	-0.018*
	(0.012)	(0.013)	(0.024)	(0.015)	(0.010)
Deposit ratio	0.147***	0.166***	0.028	0.016	0.037*
	(0.025)	(0.026)	(0.042)	(0.026)	(0.020)
Fixed Effects					
Firm	✓	✓	✓	✓	✓
County-ind.-yr.	✓	✓	✓	✓	✓
Observations	420,298	419,743	420,366	419,806	416,532

Notes: This table repeats Table 10 leaving missing values of the outcome variables missing, so the number of firm-years differs across columns. All variables, fixed effects, and the sample period are as in Table 10. The R-squared ranges from 0.41 to 0.98 across columns. Robust standard errors (clustered at the firm level) are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B Supplementary Figures

Figure B.1: **Changes in Funding-cost Gap.** This figure shows the evolution of the livret-A rate, the ECB's deposit facility (DF) rate, and the difference between them from 2008 to 2019.

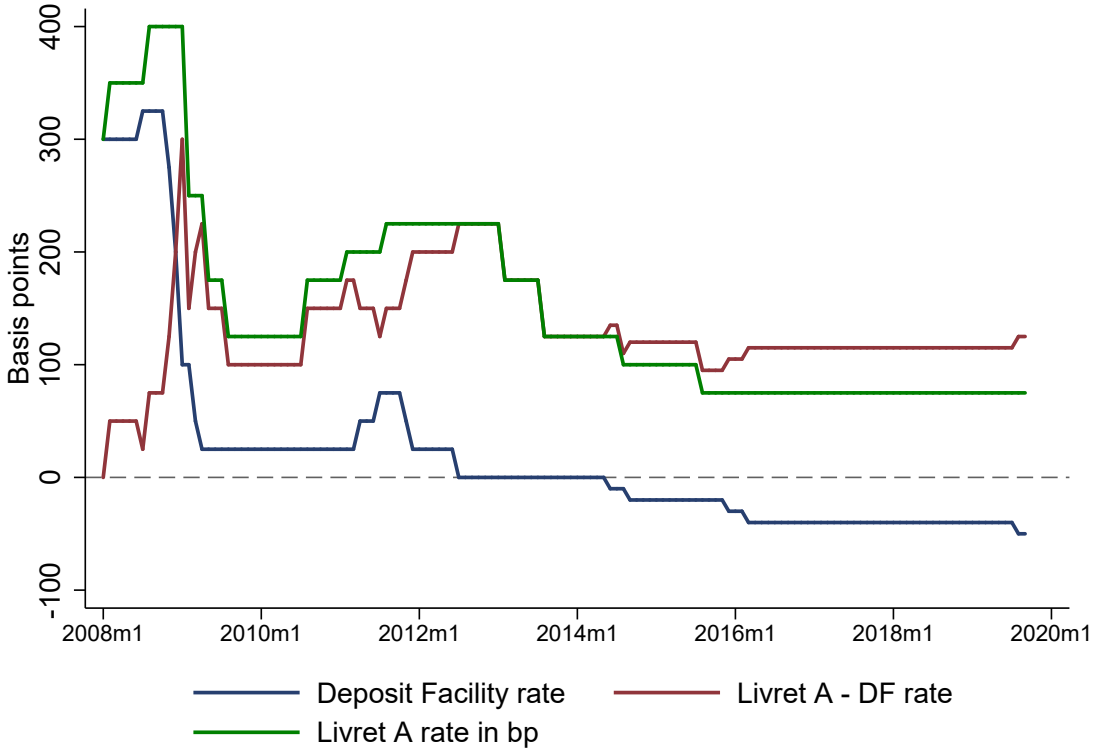
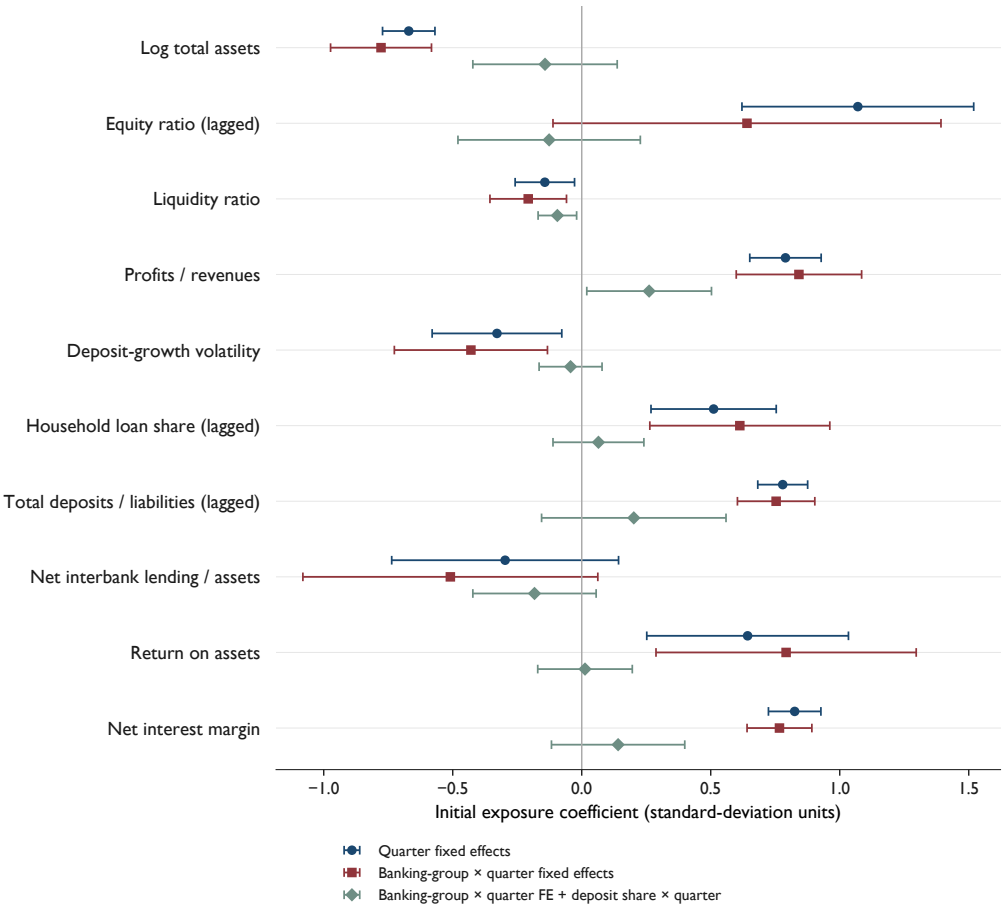


Figure B.2: **Bank Characteristics and Regulated-deposit Exposure.** This figure plots the point estimates and 95% confidence intervals from separate regressions of banks' initial regulated-deposit ratio on one bank characteristic. Circles: quarter fixed effects. Squares: banking group × quarter fixed effects. Diamonds: banking group × quarter fixed effects and initial total-deposit share × quarter fixed effects. Coefficients are scaled by the ratio of the standard deviation of the characteristic to that of the regulated-deposit ratio, so each marker is the change in the regulated-deposit ratio, in standard deviations, per standard deviation of the characteristic. Characteristics are winsorized at the 1st and 99th percentiles, regressions are weighted by initial total assets, and standard errors are clustered at the bank level. Samples range from 644 bank-quarters (139 banks) for return on assets and net interest margin to 2,726 bank-quarters (142 banks); annual accounting variables use fourth-quarter timing.



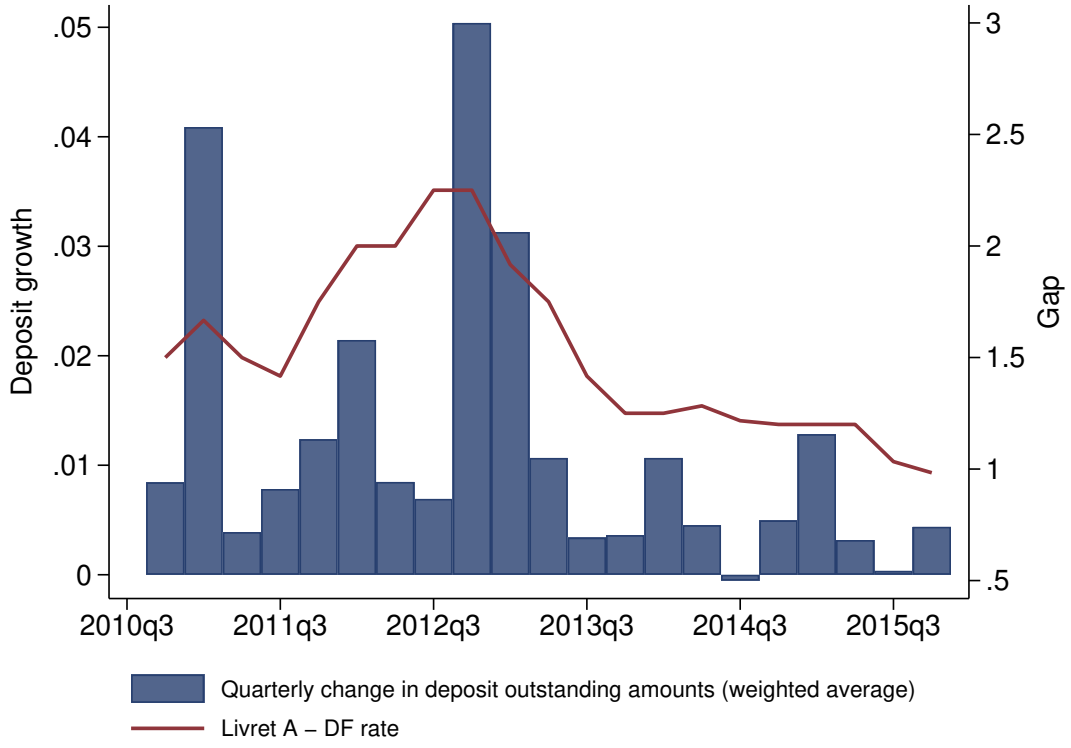
C Political Deviations from the Livret-A Formula

This appendix reproduces, in our English translation, the passages of the four press reports cited in Section 2.1.1 that describe each deviation from the livret-A formula during our sample period and the reasons given for it.

February 1, 2012. La finance pour tous, "Pas de hausse du taux du Livret A en février" (No increase in the Livret A rate in February), February 9, 2012, https://www.lafinancepourtous.com/2012/02/09/pas-de-hausse-du-taux-du-livret-a-en-fevrier/:

> "Given the price increases over the past twelve months (inflation of 2.4% year on

Figure B.3: **Sensitivity of Regulated Deposits to Funding-cost Gap.** This figure shows the quarterly growth rate in the weighted average of post-transfer regulated deposits at the bank level (accounting for entry and exit), $\frac{Deposits_{bdt} - Deposits_{bdt-1}}{0.5(Deposits_{bdt} + Deposits_{bdt-1})}$, alongside the evolution of the gap between the livret-A rate and the ECB's deposit facility rate from Q4 2010 to Q4 2015.



year in December), an increase in the Livret A rate was mathematically certain. The new rate should thus have been 2.75% instead of the current 2.25%. However, the regulations provide that the Governor of the Banque de France may depart from this automatic adjustment if he considers that exceptional circumstances justify it; the rate then remains at its previous level. In this instance, anticipating a marked decline in inflation and hence a forthcoming fall in the savings rate that could bring the savings rate back to its current level, he recommended not applying the method for calculating the upward revision of the rate of return. This recommendation was followed by the Minister of the Economy, Finance and Industry, François Baroin."

February 1, 2013. France Info, "Abaissé à 1,75%, le livret A reste avantageux" (Lowered to 1.75%, the Livret A remains attractive), January 15, 2013, https://www.franceinfo.fr/societe/le-taux-du-livret-a-sera-abaisse-a-1-75-au-1er-fevrier-annonce-le-ministre-de-l-203917.html:

"By lowering the Livret A rate by only 0.5 point, the government 'added a boost of 0.25%,' the minister said. Pierre Moscovici is playing with words somewhat, but it is true that this cut only partly reflects the slowdown in inflation. Given the calculation formula, the rate of return should normally have been lowered to 1.5%. But the minister agreed with the Governor of the Banque de France,

Christian Noyer, who had proposed bringing the Livret A rate down to 1.75%. 'There is nevertheless a concern among the French about their purchasing power,' the minister argued."

August 1, 2013. La finance pour tous, "Le livret A pourrait baisser à 1,25 %" (The Livret A could fall to 1.25%), July 12, 2013, https://www.lafinancepourtous.com/2013/07/12/le-livret-a-pourrait-baisser-a-125/:

"If one sticks to the calculation formula in place since 2009, the Livret A rate should fall on August 1 from 1.75% to 1%. Indeed, the inflation rate, the main parameter of the calculation, stood at 0.8% in June. But faced with this historically low rate, which would lead to a large change and to an unprecedented level of remuneration of the Livret A, the Governor of the Banque de France, Christian Noyer, 'decided that this exceptional circumstance justified a departure from the strict application of the rates calculated according to the formula in force, as the regulation allows.' He thus proposes that the government 'limit the damage' by ensuring savers a rate of return, net of taxes and social contributions, of 1.25%. The Minister of the Economy, Pierre Moscovici, is to have the final say very shortly."

February 1, 2014. La finance pour tous, "Le livret A maintenu à 1,25 % au 1er février 2014" (The Livret A kept at 1.25% on February 1, 2014), January 15, 2014, https://www.lafinancepourtous.com/2014/01/15/le-livret-a-maintenu-a-125-au-1er-fevrier-2014/:

"'Given the attention that the President of the Republic and the government pay to the purchasing power of popular savings, I will keep the Livret A rate at 1.25%,' Pierre Moscovici stated on Wednesday, January 15, on LCI/Radio Classique. The Minister of Finance invoked exceptional circumstances (a very low level of inflation, +0.7% year on year) to depart from the Livret A calculation method, under which the rate should mechanically have fallen to 0.75%. Christian Noyer, the Governor of the Banque de France, proposed 1%. Bercy did not follow the recommendation."

D External Validity

To assess whether our setting is specific to France, we compare the French banking sector with those of the other 21 advanced economies in the World Bank's Global Financial Development Database (GFDD). For each country and indicator, we use the latest nonmissing observation, which falls between 2008 and 2021 depending on the country and the indicator, and we do not impute missing observations. Figure D.1 to Figure D.4 rank the countries separately on each indicator.

France is close to the median in market structure. The three largest banks hold 66.3% of commercial-bank assets in France against a median of 68.5%, which places France 13th of 22 countries (Figure D.1, Panel A). Capital and reserves amount to 6.2% of French bank assets, slightly below the median of 6.8% (Panel B).

French banks also intermediate a similar volume of funds relative to the size of the economy. Private credit by deposit money banks equals 115.4% of GDP in France against a median of 106.5%, and bank deposits equal 106.7% of GDP against a median of 104.3% (Figure D.2), so France ranks 11th of 22 and 10th of 21 countries on these two measures.

French banks lend somewhat more relative to their deposits and report more nonperforming loans than the median banking sector, but neither difference places France at the edge of the distribution. Private credit equals 108.2% of bank deposits in France against a median of 95.3%, the 7th highest ratio of 21 (Figure D.3, Panel A), and nonperforming loans equal 2.7% of gross loans against a median of 1.3%, the 6th highest share of 22 (Panel B).

French firms rely on banks at least as much as firms elsewhere. Among the 12 countries with firm-survey data, 58.2% of French firms have a loan or credit line from a formal financial institution against a median of 43.3%, and 34.3% use banks to finance working capital against a median of 35.3% (Figure D.4).

The United States is the outlier in this group. Its three largest banks hold 38.4% of assets, the lowest concentration of the 22 countries. Its private bank credit equals 54.6% of GDP and 53.9% of bank deposits, about half the French ratios of 115.4% and 108.2%, and the GFDD has no firm-survey data for it.

These comparisons suggest that our estimates apply to bank-based advanced economies more broadly. The mechanism we identify, the response of credit supply to a loss of infra-marginal profits at given funding quantities and marginal funding costs, does not depend on the institutional details of the livret A. We therefore expect a comparable response of credit supply where banks are similarly concentrated, capitalized, and central to firm financing. In market-based systems such as the United States, firms can more easily substitute toward nonbank credit, so the real effects of the same shock may be smaller.

Figure D.1: **Banking Structure.** Latest nonmissing observation for each of 22 advanced economies and each indicator; observation years appear beside country names. Countries are ranked separately in each panel. The panels show the three largest banks' assets as a share of commercial-bank assets (GFDD.OI.01) and bank capital and reserves relative to total assets (GFDD.SI.03). Concentration uses the bank-level accounting sample, not consolidated banking-group shares. Capital definitions can differ across national accounting and supervisory systems. Source: World Bank, Global Financial Development Database. Dates and country coverage differ across indicators; missing observations are not imputed.

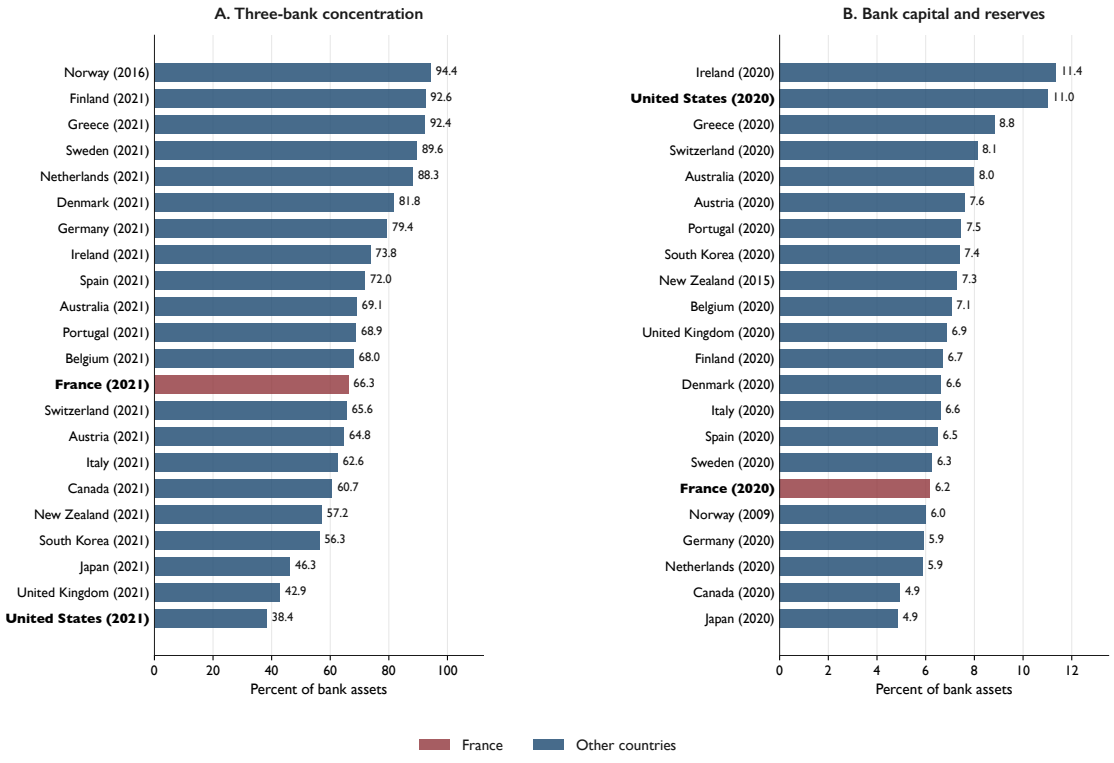


Figure D.2: **Bank Credit and Deposit Depth.** Latest nonmissing observation for each of 22 advanced economies and each indicator; observation years appear beside country names. Countries are ranked separately in each panel. The panels show private credit by deposit money banks (GFDD.DI.01) and bank deposits (GFDD.OI.02), each relative to GDP. Private credit includes household credit. The United Kingdom has no available deposit observation. Source: World Bank, Global Financial Development Database. Dates and country coverage differ across indicators; missing observations are not imputed.

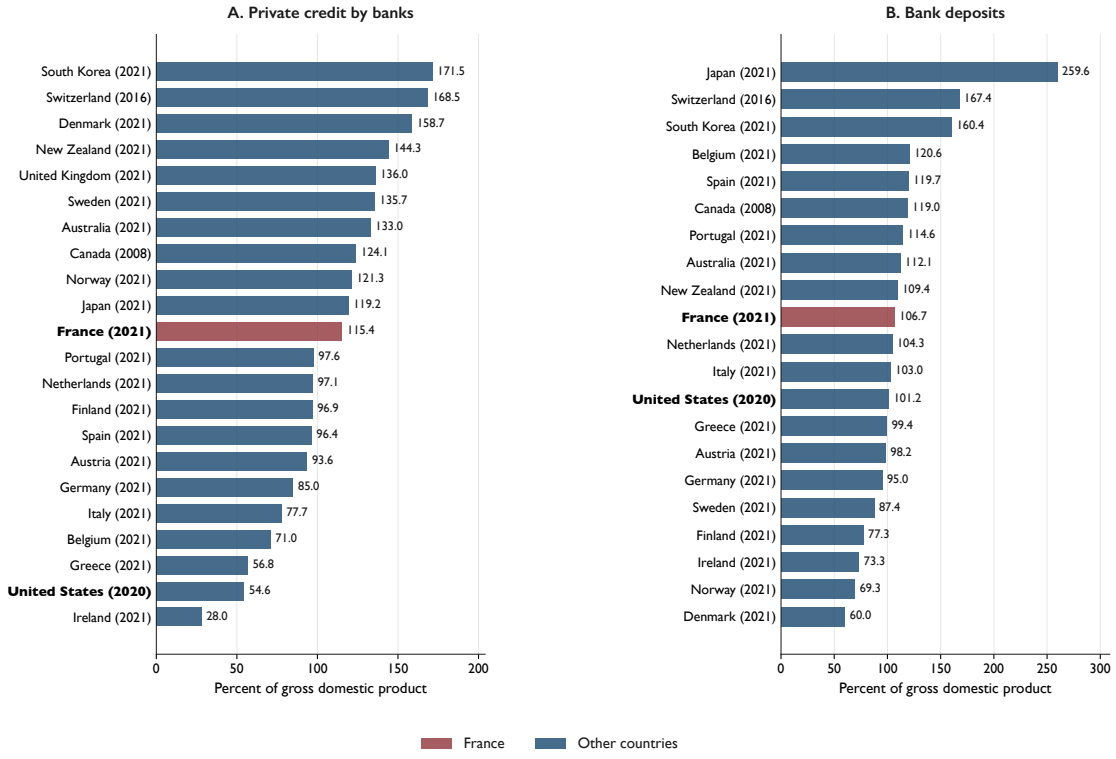


Figure D.3: **Bank Balance-sheet Indicators.** Latest nonmissing observation for each of 22 advanced economies and each indicator; observation years appear beside country names. Countries are ranked separately in each panel. The panels show bank credit relative to bank deposits (GFDD.SI.04) and nonperforming loans relative to gross loans (GFDD.SI.02). Nonperforming-loan definitions can differ across national accounting and supervisory systems. The United Kingdom has no available credit-to-deposit observation. Source: World Bank, Global Financial Development Database. Dates and country coverage differ across indicators; missing observations are not imputed.

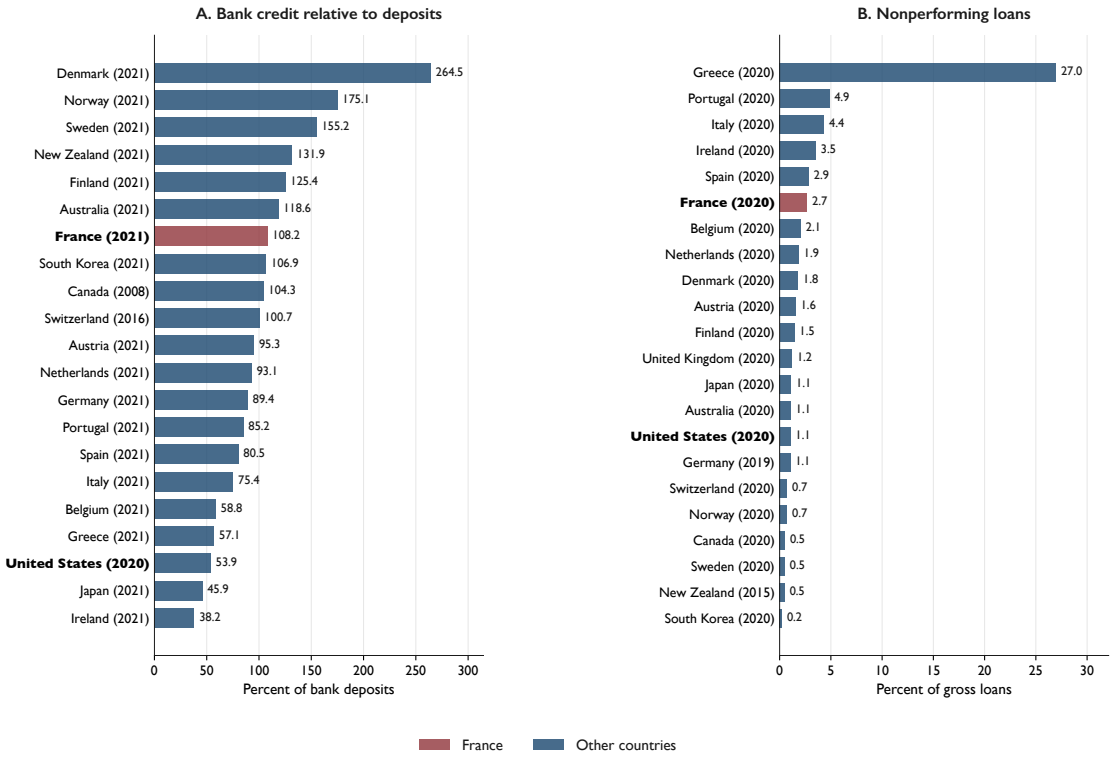


Figure D.4: **Firm Use of Financial Institutions and Banks.** Latest nonmissing observation for each of 22 advanced economies and each indicator; observation years appear beside country names. Countries are ranked separately in each panel. The panels show the share of firms with a loan or credit line from a formal financial institution (GFDD.AI.03) and the share of firms using banks to finance working capital (GFDD.AI.29). These are shares of firms, not shares of financing. Only the 12 countries with observations are shown. Source: World Bank, Global Financial Development Database. Dates and country coverage differ across indicators; missing observations are not imputed.

